STRATA: Small categories, stratified homotopy types, and constructible coefficients

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Conventions and scope

We work with nested Grothendieck universes $\mathbb{U} \in \mathbb{V}$. Small means $\mathbb{U}$-small, **Cat** denotes small categories, and categories of all small diagrams are formed in $\mathbb{V}$. We write $BS = |NS|$, $\mathcal{S}$ for the ∞-category of small spaces, and $W_\infty = \{u \mid Bu \text{ is a weak homotopy equivalence}\}$. We put $\mathrm{Hot} = \mathbf{Cat}[W_\infty^{-1}]$. An unqualified equivalence of ∞-categories is a categorical equivalence, not merely a weak equivalence of their groupoid completions. All limits and colimits of spaces or ∞-categories are derived limits and colimits. The symbol Ho is applied after forming a coherent functor category.

0.1 (Proved). The variance convention. *Throughout this paper,*

$$\mathcal{P}(S) = \mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{S}), \qquad \mathrm{HOT}(S) = \mathrm{Ho}\,\mathcal{P}(S), \qquad \mathbb{D}_{\mathcal{C}}(S) = \mathrm{Ho}\,\mathrm{Fun}(NS^{\mathrm{op}}, \mathcal{C}).$$

A sieve is closed under sources and is called open. For a poset P, its topology $P_\downarrow$ has downward closed opens. Our geometric model for this convention is the minimum-vertex *stratification of $|NP|$, equivalently the maximum-vertex stratification of $|N(P^{\mathrm{op}})|$ over P^{op} with upward closed opens. Sheaves are covariant on exit paths; therefore the finite-poset comparison identifies its exit category with P^{op}. The same abstract indexing convention applies for arbitrary P, with geometric qualifications stated in A.4–A.6.*

Proof. The principal open $U_p = P_{\leq p}$ is contained in U_q when $p \leq q$. Restriction of sections therefore gives a map from the value at q to the value at p. Reversing a simplex identifies $|NP|$ homeomorphically with $|N(P^{\mathrm{op}})|$ and changes minimum into maximum. This fixes all the oppositions used below. □

For $u : A \to B$, the objects of A/b are $(a, u(a) \to b)$ and those of $b\backslash A$ are $(a, b \to u(a))$. An ordinary final functor means that all $b\backslash A$ are nonempty and connected; *homotopy final* means that their nerves are weakly contractible. These conditions are not interchanged. For a presheaf derivator a natural transformation $\alpha : u \Rightarrow v$ induces $\alpha^* : v^* \Rightarrow u^*$. This is the index-opposite of the usual covariant derivator convention. An abstract derivator is assumed to satisfy Der 1–4 on all small categories; strongness is an additional hypothesis whenever used.

We use the relative fundamental-localizer axiom: if $A \xrightarrow{u} B \xrightarrow{v} C$ is a triangle and $A/c \to B/c$ belongs to W for every $c \in C$, then $u \in W$. Together with this we require identities, two-out-of-three, the weak retract axiom ($ri = 1$ and $ir \in W$ imply $i, r \in W$), and $A \to e \in W$ whenever A has a terminal object. This is the relative version printed as Loc (3 bis) in Chapter XVI. We do not assume that it follows from the absolute version printed there as Loc (3).

The four status labels have literal meanings. *Known* cites a theorem whose statement and required hypotheses were inspected. *Proved* records a proof here and makes no priority claim. *Refuted* includes a counterexample to the stated formulation. *Open* records what this audit has not established; it does not assert that the question is open in the literature. For geometric exodromy we distinguish ordinary constructible sheaves, constructible hypersheaves, and coefficients defined intrinsically on an abstract stratified homotopy type. An arbitrary derivator is not silently replaced by an ∞-categorical enhancement.

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A The dictionary for posets

A.1 Alexandrov sheaves and hypercompletion

A.1 (Known). The exact Alexandrov comparison. *For every small poset P, right Kan extension from principal opens gives an equivalence*

$$R_P : \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{S}) \xrightarrow{\simeq} \mathrm{Sh}(P_\downarrow)^\wedge, \qquad R_P(F)(U) = \lim_{p \in U^{\mathrm{op}}} F(p). \tag{A.1}$$

The superscript $\wedge$ denotes hypercompletion. The composite into $\mathrm{Sh}(P_\downarrow)$ is fully faithful. It is an equivalence without hypercompletion if P is finite, or more generally if every principal downset $P_{\leq p}$ is finite. For set-valued sheaves the comparison is an equivalence for every P.

Proof. The functor $p \mapsto U_p$ from P to $\mathrm{Open}(P_\downarrow)$ is fully faithful. Its induced Grothendieck topology is trivial: a covering family of U_p contains an open containing p, and any such open contains all of U_p. The basis theorem for hypercomplete sheaves (Barwick, Glasman, & Haine, 2020, Proposition 3.12.11 and Example 3.12.15) consequently identifies hypersheaves with arbitrary diagrams on these principal opens. Right Kan extension has the displayed formula. Since p is initial in $(P_{\leq p})^{\mathrm{op}}$, its value on U_p is $F(p)$; hence restriction is an inverse. Presheaf ∞-topoi are hypercomplete, since limits and truncations are objectwise. This also proves full faithfulness into ordinary sheaves.

A finite Alexandrov space has a hypercomplete sheaf ∞-topos (Barwick et al., 2020, Example 3.12.15). If each U_p is finite, restriction to this open cover detects equivalences and hypercompleteness, giving the stated extension. The ordinary comparison theorem for a basis gives the set-valued assertion without a finiteness assumption. The empty poset gives the terminal category on both sides; disjoint unions give products. □

A.2 (Refuted). Ordinary sheaves for arbitrary posets. *The removal of hypercompletion from A.1 for arbitrary posets is false. It is still false if "locally finite poset" means only that every closed order interval is finite.*

Proof. Let $Q = \mathbb{N} \times \{0, 1\}$, with $(i, a) < (j, b)$ exactly when $i < j$, and put $P = Q^{\mathrm{op}}$. Every interval is finite. Every upper open of Q is generated by at most two minimal elements, so finite gluing shows that filtered colimits in $\mathrm{Sh}(Q_\uparrow)$ are computed sectionwise. Its terminal object is therefore compact.

Set $Q_n = \{0, \ldots, n\} \times \{0, 1\}$ and $F_n = \mathrm{Lan}_{Q_n \hookrightarrow Q}(*)$. Pointwise, $\mathrm{colim}_n F_n = *$, whereas, at rank greater than n,

$$F_n(j, b) = BQ_n = (S^0)^{*(n+1)} \simeq S^n.$$

If $*$ were compact in $\mathrm{Fun}(Q, \mathcal{S})$, it would be a retract of some F_n. The essential image of this fully faithful left Kan extension is closed under retracts, by its counit criterion. Thus $*$ would itself be that left Kan extension, contradicting $S^n \not\simeq *$. The terminal objects have different compactness properties, so the categories cannot be equivalent. This is the counterexample of Aoki (2023, Example A.13), with the variance reversed. □

A.3 (Proved). Presentable coefficients on the Alexandrov model. *Let $\mathcal{C}$ be a presentable ∞-category. Define $\mathrm{Sh}^\wedge(P_\downarrow; \mathcal{C})$ by hyperdescent, equivalently by testing hyperdescent after every functor $\mathrm{Map}_{\mathcal{C}}(c, -)$. Then*

$$\mathrm{Sh}^\wedge(P_\downarrow; \mathcal{C}) \simeq \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{C}),$$

with the same limit formula as A.1. If every principal downset is finite, ordinary $\mathcal{C}$-valued sheaves give the same category.

Proof. The presheaf Yoneda embedding $\mathcal{C} \to \mathrm{Fun}(\mathcal{C}^{\mathrm{op}}, \mathcal{S})$ preserves and detects limits. Apply the basis comparison of A.1 after $\mathrm{Map}_{\mathcal{C}}(c, -)$, for every c, to obtain the right Kan extension formula and its inverse. The diagrams so obtained are represented in $\mathcal{C}$ because the requisite small limits exist. Hyperdescent and the unit and counit equivalences are detected by these mapping-space functors. Under the finiteness hypothesis, a space-valued sheaf on each principal open is already hypercomplete; applying this to all these mapping-space functors proves the last assertion. This argument uses no compactness assumption on c. □

A.2 Realization and the geometric qualification

A.4 (Proved). The minimum-vertex stratification. *For a finite poset P let $Z_P = |NP|$ and let*

$$s_P : Z_P \longrightarrow P_{\downarrow}, \qquad s_P([p_0 < \cdots < p_k; t_0, \ldots, t_k]) = \min\{p_i : t_i > 0\}.$$

This is a continuous stratification. Each stratum is contractible, and its exit-path ∞-category is equivalent to P^{op}. Consequently

$$\mathrm{Sh}_{P\text{-}cons}(Z_P; \mathcal{S}) \simeq \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{S}) \simeq \mathrm{Sh}(P_{\downarrow}; \mathcal{S}). \tag{A.4}$$

The statement also holds locally when the order complex is locally finite. Here strata are allowed to be locally contractible polyhedra; they are not required to be manifolds.

Proof. For a downset U, the inverse image $s_P^{-1}U$ consists of points with positive barycentric coordinate at some vertex in U, and is a union of open vertex stars. The stratum of p is the part of the cone $\{p\} * |N(P_{>p})|$ on which the coordinate of p is positive. Moving this coordinate to one contracts the stratum to p.

We also verify conicality of this particular coarsening. Write $C_\infty T$ for the open cone with radial coordinate in $[0, \infty)$, including its apex when $T = \varnothing$. On the open vertex star $\mathrm{St}^\circ(p) = \{t_p > 0\}$, divide all other barycentric coordinates by t_p. The coordinates below and above p vary independently, since a chain below p and a chain above p concatenate with p to a chain. This gives a homeomorphism

$$\mathrm{St}^\circ(p) \cong C_\infty |N(P_{<p})| \times C_\infty |N(P_{>p})|. \tag{A.4a}$$

Explicitly, the two radial coordinates are $r_- = \sum_{q<p} t_q/t_p$ and $r_+ = \sum_{q>p} t_q/t_p$; their angular coordinates are the respective normalized barycentric coordinates. The inverse has $t_p = (1 + r_- + r_+)^{-1}$ and rescales each angular coordinate by $r_\pm/(1 + r_- + r_+)$. These formulas respect all face identifications, and are continuous in the cone topologies. The stratum of p is the first apex times the second cone. Away from the first apex the stratum is determined by the minimum in the lower link, independently of the second factor. Thus (A.4a), with the factors interchanged, is a conical chart over P^{op} in the usual upward-open convention. The stratum factor is a locally contractible polyhedron. Consequently the raw exit simplicial set is an ∞-category (Lurie, 2017, Theorem A.6.4), and agrees with the intrinsic exit category of this exodromic stratification (Haine, Porta, & Teyssier, 2026, Recollection 5.6.1).

We give the categorical comparison explicitly. Let $\mathsf{Ch}(P)$ be the face poset of nonempty finite chains, with arrows given by inclusion. The cell stratification of Z_P has exit category $\mathsf{Ch}(P)$: an exit from an open face to an incident coface has a contractible space of choices in a linear simplex, and the simplexwise choices and their compositions glue. This is the simplicial exit calculation (Lurie, 2017, Theorem A.6.10). Passing to the coarser stratification localizes this category at inclusions preserving the minimum. The localization functor is

$$m : \mathsf{Ch}(P) \to P^{\mathrm{op}}, \qquad \sigma \mapsto \min \sigma.$$

Apply the universally localizing last-vertex map to $N(P^{\mathrm{op}})$ (Kerodon, 2026a, Proposition 6.3.6.2). Its subdivision is $N\mathsf{Ch}(P)$, and its last vertex is the minimum in P. This proves the asserted localization including its mapping spaces. Explicitly, an arrow $p \to q$ in P^{op}, with $q \leq p$, is represented by $\{p\} \to \{q, p\} \leftarrow \{q\}$, where the backward arrow is inverted. For $r \leq q \leq p$, the chain $\{r, q, p\}$ supplies the composition relation. Every chain is identified with its minimum by $\{\min \sigma\} \to \sigma$; the universal localization theorem supplies the full coherence assertion beyond these elementary roofs.

The coarsening theorem for exodromy identifies this localization with the intrinsic exit category of s_P (Haine et al., 2026, Theorem 3.3.5 and Theorem 5.1.7); the conical charts just proved identify it with the raw exit category as well. The finite polyhedron is paracompact and locally of singular shape, and the finite stratifying poset satisfies the ascending chain condition. Hence Lurie (2017, Proposition A.5.9 and Theorem A.9.3) give (A.4) for ordinary constructible sheaves and show that these sheaves are hypercomplete. For a locally finite order complex the argument applies on finite star neighbourhoods and glues by open descent. No assertion about an arbitrary infinite-dimensional realization is hidden in this last local argument. □

A.5 (Known). The coefficient range of geometric exodromy. *For the finite geometric model in A.4, (A.4) holds with $\mathcal{S}$ replaced by any compactly assembled presentable ∞-category $\mathcal{C}$, and also by any stable presentable ∞-category. More generally, for every $\mathcal{C}$ there is a fully faithful functor*

$$\mathrm{Fun}(P^{\mathrm{op}}, \mathcal{C}) \simeq \mathrm{Sh}_{P\text{-}cons}(Z_P; \mathcal{S}) \otimes \mathcal{C} \hookrightarrow \mathrm{Sh}_{P\text{-}cons}(Z_P; \mathcal{C}). \tag{A.5}$$

For an arbitrary presentable $\mathcal{C}$, its essential surjectivity is an additional assertion, not a consequence of presentability alone.

Proof. The inclusion of constructible space-valued sheaves in all sheaves is both a left and a right adjoint for this exodromic stratification. Tensoring with $\mathcal{C}$ preserves its fully faithful adjunction and places the image in the constructible objects. The identification of the tensor product with the functor category follows from Lurie (2017, Proposition 4.8.1.17). The essential-surjectivity assertions are precisely Haine et al. (2026, Lemma 4.1.2 and Corollary 4.1.15): compactly assembled coefficients work in general, while stable presentable coefficients work for a noetherian stratifying poset. Our poset is finite. □

A.6 (Open). The unrestricted geometric clauses. *This audit does not establish that (A.5) is essentially surjective for every presentable $\mathcal{C}$, even for every finite poset P. Nor does it establish, for an arbitrary small poset with its unrestricted weak CW realization, that ordinary constructible sheaves on Z_P are precisely* $\mathrm{Fun}(P^{\mathrm{op}}, \mathcal{S})$. *These are separate statements from the fully proved Alexandrov hypersheaf assertion A.1.*

The attempted tensor argument stops at the fully faithful functor (A.5): tensoring does not by itself identify all objects detected as locally constant after restriction to strata. For the unrestricted realization, the finite-star proof stops when such a star is no longer finite; ordinary and hyperconstructibility can differ. The first coefficient instance not settled by that argument is the two-element poset with a presentable coefficient category outside the coefficient classes of A.5. The explicit infinite poset in A.2 is the first realization test retained here; A.2 refutes its ordinary Alexandrov clause, not by itself its geometric clause. A.12 nevertheless computes the raw exit category for every small poset; the unproved assertion here concerns its classification of ordinary constructible sheaves. We have not constructed a counterexample to either assertion in this paragraph.

A.3 Restriction, extensions, and local systems

A.7 (Proved). The three functors and the comma formulas. *For a monotone map $u : P \to Q$ and presentable $\mathcal{C}$, restriction has both adjoints,*

$$u_! \dashv u^* \dashv u_* : \mathrm{Fun}(P^{\mathrm{op}}, \mathcal{C}) \rightleftarrows \mathrm{Fun}(Q^{\mathrm{op}}, \mathcal{C}),$$

where the displayed arrows denote the three appropriately directed functors and

$$(u_! F)(q) = \operatorname*{colim}_{(q\backslash P)^{\mathrm{op}}} F, \qquad (u_* F)(q) = \lim_{(P/q)^{\mathrm{op}}} F. \tag{A.7}$$

They are the corresponding adjoints on Alexandrov hypersheaves. On the finite geometric models, in the coefficient ranges of A.5, restriction is geometric inverse image. For arbitrary $\mathcal{C}$ the same compatibility holds on the fully faithful diagram images of (A.5). The other two functors are the intrinsic adjoints on these diagram categories and, when A.5 is an equivalence, on the whole constructible categories. They are not, without further hypotheses, the ambient topological pushforwards.

Proof. These are the pointwise formulas for left and right Kan extension along u^{op}. An arrow $u^{\mathrm{op}}(p) \to q$ in Q^{op} is an arrow $q \to u(p)$ in Q, giving the first index; the second is dual. For hypersheaves the functors are transported through A.3. For finite Alexandrov spaces, inverse image takes stalk q to stalk $u(p)$, and direct image has sections over $u^{-1}(U_q) = \{p : u(p) \leq q\}$, giving the second formula directly. In the geometric model $s_Q|Nu| = us_P$; the functor on exit categories is u^{op}, so exodromy identifies inverse image with restriction. The uniqueness of adjoints then identifies its intrinsic adjoints, but says nothing about preservation of constructibility by ambient operations. Part D gives a finite counterexample to that additional claim.

For $P = \varnothing$, the first formula gives an initial object and the second a terminal object at every q. These formulas remain valid when u misses connected components. The symbol $u_!$ here means left Kan extension, not a prior compact-support construction on topological spaces. □

A.8 (Proved). Strict fibres versus comma categories. *For $q \in Q$, the strict fibre $P_q = \{p : u(p) = q\}$ has canonical full inclusions*

$$k_q : P_q \hookrightarrow q\backslash P, \qquad l_q : P_q \hookrightarrow P/q.$$

The comparison $\mathrm{colim}_{P_q^{\mathrm{op}}} F \to (u_! F)(q)$ is an equivalence for every diagram on the comma category whenever k_q^{op} is homotopy final. The comparison $(u_* F)(q) \to \lim_{P_q^{\mathrm{op}}} F$ is an equivalence for every diagram whenever l_q^{op} is homotopy initial. The precise tests are, respectively, weak contractibility of

$$\{x \leq p : u(x) = q\} \quad (q \leq u(p)), \qquad \{x \geq p : u(x) = q\} \quad (u(p) \leq q). \tag{A.8}$$

Proof. For a point $(p, q \leq u(p))$ of $q\backslash P$, the comma category testing finality of k_q^{op} has objects $x \in P_q$ with $x \leq p$. Its nerve is the nerve of the first displayed poset, possibly with opposition, which does not change weak contractibility. For limits the initiality test gives the second poset in the same way. The homotopy cofinality theorem now gives the comparisons, including their directions. Empty comma categories fail the test. Testing only the total homotopy type of k_q or l_q would be insufficient for all coefficients; the pointwise comma tests are what is used. Part D identifies them with smoothness and properness under the manuscript's variance. □

A.9 (Known). Local systems and the reflection. *Let S be any small category and let $\ell : NS^{\mathrm{op}} \to BS$ be groupoid completion, using reversal to identify the completions of S and S^{op}. For every presentable $\mathcal{C}$, restriction is fully faithful and has both adjoints:*

$$\ell_! \dashv \ell^* \dashv \ell_*.$$

Its image is exactly the diagrams taking every arrow to an equivalence. Thus

$$\mathcal{P}(S)^{\mathrm{lc}} \simeq \mathrm{Fun}(BS, \mathcal{S}), \qquad \mathrm{HOT}^{\mathrm{lc}}(S) \simeq \mathrm{Ho}\,\mathrm{Fun}(BS, \mathcal{S}). \tag{A.9}$$

The reflection of VI ms. 65–69 is, by its universal property, $\rho = \ell_!$ with inclusion ℓ^*; its endofunctor on all diagrams is $\ell^*\ell_!$. The right projector is $\ell^*\ell_*$.

Proof. The universal property of ℓ identifies functors on BS with exactly the functors inverting all arrows of NS^{op}, including their mapping spaces. Small limits and colimits in $\mathcal{C}$ give both Kan extensions. The counit $\ell_!\ell^* \to 1$ and the unit $1 \to \ell_*\ell^*$ are equivalences because ℓ^* is fully faithful. The manuscript's reflector is characterized by the same adjunction and is therefore uniquely equivalent as an adjoint, not literally equal as a representative construction. Cisinski proves the corresponding assertion for arbitrary derivators (Cisinski, 2009, Theorem 3.13); the attached GALOIS A.6 and A.8 state these comparisons with the same variance.

For the downset Alexandrov model this is local *hyper*constancy. In the finite setting it is ordinary local constancy: constancy on U_q forces $F(q) \to F(p)$ to be an equivalence for every $p \leq q$, and conversely all such equivalences identify the restriction with a constant diagram on U_q. The same interpretation on Z_P follows from (A.4). We do not identify local hyperconstancy with ordinary local constancy on every infinite-dimensional space. □

A.10 (Proved). A computation and the need for homotopy colimits. *For $P = [1]$, write a presheaf as $F = (B \xrightarrow{v} A)$. Its reflection and coreflection onto local systems are the constant arrows on A and B, respectively. An ordinary colimit cannot replace a homotopy colimit in the preceding formulas without an additional hypothesis on the diagram or its indexing category.*

Proof. The category $[1]^{\mathrm{op}}$ has terminal object 0 and initial object 1, so its colimit is A and its limit is B. The unit of the reflection has components 1_A and v; the counit of the coreflection has components v and 1_B.

For the second assertion, let I be the face poset of the boundary of a triangle and use the constant simplicial-set diagram with value a point. Its strict colimit is a point, while its homotopy colimit is $BI \cong S^1$. The category has six objects, the six identities and six vertex–edge incidence arrows, and no composable pair of nonidentity arrows. The normalized nerve chain calculation in `code/AB_checks.py` gives $H_0 = \mathbb{Z}$, $H_1 = \mathbb{Z}$, and no higher homology. This explicitly prevents the proposed replacement. □

A.11 (Proved). What restriction of an abstract derivator supplies. *For a derivator $\mathbb{D}$ in presheaf variance, $\mathbb{D}(P)$, its restriction functors, both Kan extensions, and their pointwise comma formulas give a well-defined theory of cellular coefficients on posets. The locally constant subderivator has both adjoints as in A.9. If $\mathbb{D} = \mathbb{D}_{\mathcal{C}}$ and the geometric coefficient hypotheses of A.5 hold, this theory is the homotopy category, at every diagram shape, of constructible $\mathcal{C}$-valued sheaves. Without an enhancement or a separately defined sheaf theory, the phrase "$\mathbb{D}$-valued topological sheaf" is not an additional consequence of Der 1–4.*

Proof. The first assertions are Der 3–4 and Cisinski (2009, Theorem 3.13). In the enhanced case, apply the exodromy equivalence before taking homotopy categories and with an additional indexing shape; the resulting restrictions and adjoints agree by their universal properties. The final assertion concerns the data available: Der 1–4 specify categories of coherent indexed diagrams and Kan extensions, but do not specify a topology-valued sheaf construction or mapping spaces. Defining cellular $\mathbb{D}$-coefficients by $\mathbb{D}(P)$ is consistent and useful, but cannot prove an equivalence with an unspecified category of topological sheaves. □

A.4 The unrestricted raw exit calculation

A.12 (Proved). Raw exit paths of an arbitrary poset realization. *For every small poset P, the minimum-vertex map $s_P : |NP| \to P_\downarrow$ is continuous. Its raw exit simplicial set, formed using the upward-open target P^{op}, is an ∞-category, and the projection recording the stratum labels is a categorical equivalence*

$$\mathrm{Sing}^{P^{\mathrm{op}}}(|NP|, s_P) \longrightarrow N(P^{\mathrm{op}}). \tag{A.12}$$

Here $|NP|$ has its weak CW topology. This statement does not identify ordinary constructible sheaves on that unrestricted realization.

Proof. Each barycentric coordinate is continuous in the weak CW topology, because its restriction to every closed simplex is continuous. The inverse image of a downset is the union of the open vertex stars at its elements, which proves continuity.

We recall the compactness property of a CW complex in the form needed here. A compact subset meets only finitely many open cells. Indeed, otherwise choose one point in each of countably many distinct open cells. Every closed cell contains points from only finitely many of these open cells, by closure finiteness. Consequently every subset of the chosen point set has closed intersection with every closed cell, and is closed in the CW complex by its weak topology. The point set would therefore be an infinite closed discrete subset of a compact space, a contradiction. Taking the closures of the finitely many cells met by a compact subset places that subset in a finite subcomplex. This is also the finite-subcomplex reduction used in Lurie (2017, Corollary A.6.9).

Let P_0 run through the finite subsets of P, with their induced orders. A finite subcomplex of $|NP|$ is contained in $|NP_0|$ for some such P_0: take all vertices of its finitely many cells. Every singular exit simplex has compact image and hence factors through $|NP_0|$. Its stratum labels also belong to P_0, since every vertex of the domain simplex has the prescribed label of its image. Thus there is an equality of simplicial sets

$$\mathrm{Sing}^{P^{\mathrm{op}}}(|NP|, s_P) = \operatorname*{colim}_{P_0 \subset P,\, |P_0| < \infty} \mathrm{Sing}^{P_0^{\mathrm{op}}}(|NP_0|, s_{P_0}). \tag{A.12a}$$

The transition maps are monomorphisms. Any inner horn in this filtered union occurs at a finite stage, where A.4 supplies a filler. Hence the union is an ∞-category.

For completeness, the same finite-data observation verifies that the union of the projections in A.4 is a categorical equivalence. Any two objects occur together at a finite stage. The right mapping simplicial sets between them are filtered unions of the corresponding right mapping simplicial sets at these finite stages: their n-simplices and defining endpoint conditions use only the finite simplex Δ^{n+1}. These are Kan complexes, and their filtered unions compute their filtered colimits as spaces, since each horn, sphere, and homotopy uses only finitely many simplices. A.4 identifies each stage with the appropriate mapping space in $N(P_0^{\mathrm{op}})$. Thus the resulting mapping space is contractible if the two stratum labels p, q satisfy $q \leq p$, and

is empty otherwise. Every label p occurs at the vertex p, so the projection is also essentially surjective. This proves (A.12), including mapping spaces. The argument concerns finite domains of singular simplices; it does not imply that a constructible sheaf is recovered from these simplices without an exodromy theorem. □

B General small categories and the two invariants

B.1 EI categories and conservative stratifications

B.1 (Proved). Exactly which ordinary categories are layered. *A small ordinary category S admits a conservative functor to a poset if and only if every endomorphism in S is invertible. In this EI case, let P_S be its set of isomorphism classes, ordered by*

$$[s] \leq [t] \quad \Longleftrightarrow \quad S(s,t) \neq \varnothing.$$

This is a poset, the quotient $q : S \to P_S$ is conservative, and the fibre over $[s]$ is a connected groupoid equivalent to the one-object groupoid $\mathrm{Aut}_S(s)$. Thus $N(S^{\mathrm{op}}) \to P_S^{\mathrm{op}}$ is an abstract stratified homotopy type, with stratum $B\,\mathrm{Aut}_S(s)$.

Proof. A conservative functor reflects isomorphisms. Since a poset has no nonidentity endomorphism, it forces every endomorphism of its source to be invertible. Conversely suppose S is EI. Reflexivity and transitivity of the displayed relation follow from identities and composition. If $f : s \to t$ and $g : t \to s$ exist, then gf and fg are invertible. The morphism $(gf)^{-1}g$ is a left inverse to f, and $g(fg)^{-1}$ is a right inverse, so they coincide and f is invertible. This proves antisymmetry on isomorphism classes. An arrow between isomorphic objects is invertible by the same argument, proving conservativity and the fibre description. Choosing one object in that fibre gives a fully faithful, essentially surjective functor from its automorphism groupoid.

The passage from a conservative functor to a poset to an abstract stratified homotopy type is the layered-category formulation of the stratified homotopy hypothesis (Barwick et al., 2020, §§2.1,2.3). Here it is used as the definition of the abstract object; its underlying category and its conservativity have just been proved. Ordinary EI categories give the cases with discrete mapping spaces. General stratified homotopy types have mapping spaces of arbitrary homotopy dimension and therefore extend beyond this ordinary subcategory. □

B.2 (Proved). The exact abstract EI coefficient dictionary. *For the EI category in B.1, its intrinsic constructible coefficient category is*

$$\mathrm{Cons}_{\mathcal{C}}(S) = \mathrm{Fun}(S^{\mathrm{op}}, \mathcal{C}).$$

On the stratum $[s]$ this restricts to $\mathrm{Fun}(B\,\mathrm{Aut}_S(s), \mathcal{C})$. For space-valued coefficients, these categories are the strata of the presheaf ∞-topos $\mathcal{P}(S)$ stratified over P_S^{op}. On any exodromic realization whose exit category is $N(S^{\mathrm{op}})$, intrinsic coefficients coincide with geometric constructible coefficients in the coefficient range of exodromy.

Proof. The functor $q^{\mathrm{op}} : S^{\mathrm{op}} \to P_S^{\mathrm{op}}$ induces precomposition between presheaf ∞-topoi. It preserves finite limits and all colimits, so supplies the required inverse-image functor of the stratification. Restriction to the full subgroupoid over $[s]$ gives exactly the displayed stratum category. More concretely, lower ideals of P_S are sieves in S, and their complements are cosieves; restriction and extension by the initial or terminal presheaf give the open and closed subtopoi. Isolating $[s]$ as an open in its closure gives the same full-groupoid restriction.

Groupoid inversion identifies the two variances on a stratum, so $\mathrm{Fun}(B\,\mathrm{Aut}_S(s), \mathcal{C})$ has the stated type.

This construction retains the hom-sets $S(s,t)$, their left and right automorphism actions, and composition. It does not replace them by the mere assertion that $[s] \leq [t]$. The equality of intrinsic coefficients is their definition on the specified stratified category, not a claim that an arbitrary topology realizes them. When geometric exodromy is available, its equivalence is covariant on exit paths, giving precisely $\mathrm{Fun}(N(S^{\mathrm{op}}), \mathcal{C})$. Part A states the distinction between this conditional geometric conclusion and the unconditional categorical one. □

B.3 (Proved). A literal geometric dictionary for finite acyclic categories. *Let S be a finite category whose only endomorphisms are identities and which has no isomorphisms between distinct objects. The minimum-object map on $|NS|$, with target P_S and its downset topology, defines an exodromic stratification with contractible strata and exit category $N(S^{\mathrm{op}})$. In particular*

$$\mathrm{Sh}_{P_S\text{-}cons}(|NS|; \mathcal{S}) \simeq \mathrm{Fun}(S^{\mathrm{op}}, \mathcal{S}).$$

The coefficient extensions in A.5 apply. Multiple parallel arrows are allowed; replacing S by P_S generally changes these coefficients.

Proof. A nondegenerate simplex of NS is a string of nonidentity arrows. Its objects are distinct, since a return to an object would produce a nonidentity endomorphism or an isomorphism between different objects. Consequently its characteristic map is injective on every face and on its whole closed simplex. The finitely many nondegenerate simplices form a finite regular cell complex, even when two different cells have the same vertex set. Its barycentric subdivision is the simplicial complex of chains in the face poset $\mathsf{F}(S)$.

With its cell stratification the intrinsic exit category is $\mathsf{F}(S)$: under the barycentric homeomorphism this is the maximum-face stratification of $|N\mathsf{F}(S)|$, so the dual convention in A.4 gives the assertion. The minimum-object map is continuous, since the inverse image of a downset is the union of the open stars of its vertices. Its stratum at s contracts to the vertex s by increasing the coordinate at s; this formula is compatible with every face identification on which the coordinate at s remains positive. Coarsening the cell stratification localizes $\mathsf{F}(S)$ at face inclusions preserving that object.

We verify that the minimum-object stratification is itself conical, so that its intrinsic and raw exit categories agree. Define the finite acyclic categories

$$L_s^- = (S/s) \setminus \{\mathrm{id}_s\}, \qquad L_s^+ = (s\backslash S) \setminus \{\mathrm{id}_s\},$$

where the displayed subcategories are full. A simplex containing s splits uniquely into a string ending at s and a string starting at s. The first string, without s, is a simplex of NL_s^-, and the second is a simplex of NL_s^+. Conversely those two strings concatenate uniquely through s. Dividing their barycentric coordinates by the positive coordinate at s, exactly as in (A.4a), therefore gives a homeomorphism

$$\mathrm{St}^\circ(s) \cong C_\infty|NL_s^-| \times C_\infty|NL_s^+|. \tag{B.3a}$$

The cone notation and its empty-link convention are those of A.4. Here the arrows into or out of s are retained as part of the link objects; in particular, parallel arrows are not identified by this formula. Concatenation and splitting commute with all face identifications that retain s, proving that the two coordinate formulas are mutually inverse continuous maps. The stratum at s is the first apex times the second cone. At a point with nonzero lower radial coordinate, its label is the first object in the lower string, independently of the upper factor. Thus (B.3a), with the factors interchanged, is a conical chart over P_S^{op}. Its stratum factor is a locally

contractible polyhedron. By Lurie (2017, Theorem A.6.4) and Haine et al. (2026, Recollection 5.6.1), the raw exit simplicial set is an ∞-category and agrees with the intrinsic exit category.

The nerve of S^{op} is nonsingular: each of its nondegenerate simplices embeds, as just checked. Its subdivision is $N\mathsf{F}(S)$, and its last-vertex map sends a simplex to its first object in S, with the appropriate composite arrow on a face inclusion. The universally localizing subdivision theorem (Kerodon, 2026a, Proposition 6.3.6.2 and Remark 6.3.6.12) identifies this localization with $N(S^{\mathrm{op}})$. The geometric coarsening theorem used in A.4 identifies it with the intrinsic exit category of the stated stratification, and (B.3a) identifies this with its raw exit category. The finite polyhedron is paracompact and locally of singular shape, while P_S^{op} satisfies the ascending chain condition. Thus Lurie (2017, Proposition A.5.9 and Theorem A.9.3) prove the ordinary constructible coefficient equivalence. This proof keeps parallel arrows as separate one-cells and their composition as the specified two-cells, rather than discarding either datum. □

B.4 (Proved). The complete projective-plane test. *Let R_2 have objects $0, 1, 2$, identity endomorphisms, no arrows from a larger object to a smaller one, and*

$$R_2(i,j) = \mathbb{Z}/2 \quad (i<j), \qquad (j \xrightarrow{b} k)(i \xrightarrow{a} j) = (i \xrightarrow{a+b} k).$$

Then BR_2 is homeomorphic to $\mathbb{RP}^2$. Under that homeomorphism its minimum-object strata are

$$U_0 = \{[x_0 : x_1 : x_2] : x_0 \neq 0\} \cong \mathbb{R}^2, \quad U_1 = \{x_0 = 0, x_1 \neq 0\} \cong \mathbb{R}, \quad U_2 = \{[0:0:1]\}.$$

Their exit category is R_2^{op}, so their space-valued constructible sheaves are exactly $\mathrm{Fun}(R_2^{\mathrm{op}}, \mathcal{S})$.

Proof. Let Q have objects (i, ϵ), $i = 0, 1, 2$ and $\epsilon \in \{+1, -1\}$, with a unique arrow $(i, \epsilon) \to (j, \delta)$ when $i < j$, and otherwise only identities. Composition is the unique arrow with the prescribed endpoints. The involution flips every sign. Its order complex is the join of three two-point sets, realized as the boundary of the octahedron with vertices $\pm e_0, \pm e_1, \pm e_2$ in $\mathbb{R}^3$. Radial projection gives an equivariant homeomorphism to S^2, and the involution becomes the antipodal involution.

The quotient functor $Q \to R_2$ sends a sign pair on an arrow to 0 if the signs agree and to 1 otherwise. Sign comparisons add modulo two, so this is a functor. Every string in R_2 has exactly two lifts to Q, related by the global sign flip: choose its initial sign, and its labels force every subsequent sign. This includes degeneracies. Consequently $NQ/(\mathbb{Z}/2) = NR_2$ degreewise, with all face and degeneracy operators respected. Realization commutes with this quotient, and the action on BQ is free, so

$$BR_2 \cong BQ/(\mathbb{Z}/2) \cong S^2/(x \sim -x) = \mathbb{RP}^2.$$

In octahedral coordinates the first nonzero coordinate is precisely the minimum level in its chain. This proves the stated description of the strata. B.3 applies, proving the exit calculation including composition. Explicitly there are two exit classes $j \to i$ for $i < j$, and their composite is the sum of their sign-change labels. Their mapping spaces have contractible components, not merely two components: this is part of the localization equivalence in B.3.

This is the $n = 2$ case of the construction recorded by Carcacía Campos (2026, Theorem 6.2). The independent checks in `AB_checks.py` enumerate the nine arrows and all compositions, verify the two-lift assertion on every nondegenerate simplex, and compute the normalized integral boundary matrices. For R_2 the simplex counts are $(3, 6, 4)$ and $H_1 = \mathbb{Z}/2$; for Q they are $(6, 12, 8)$ and $\widetilde{H}_2 = \mathbb{Z}$. The homeomorphism proof above, rather than homology alone, identifies the spaces. □

B.5 (Open). The unrestricted literal EI realization clause. *For an arbitrary small EI category, B.1–B.2 establish its abstract stratified homotopy type and intrinsic coefficient dictionary. This audit does not establish, without further geometric hypotheses, that ordinary constructible sheaves on the particular unrestricted weak CW realization $|NS|$ with its minimum-isomorphism-class stratification always give $\mathcal{P}(S)$.*

The finite acyclic proof uses a finite regular cell refinement. Nontrivial automorphisms produce simplices in arbitrarily high dimensions, and an arbitrary infinite category can have nonfinite stars. The missing step is geometric exodromy, including the distinction between sheaves and hypersheaves, for that prescribed realization. Conservativity of $N(S^{\mathrm{op}}) \to P_S^{\mathrm{op}}$ does not by itself prove that step. A groupoid with one stratum is affirmative by ordinary local-system classification on its locally contractible classifying space. Among posets, the unbounded layered example A.2 remains an explicit unsettled realization test; among finite acyclic categories there is no remaining case. The first mixed finite EI candidate outside that proof has two objects $0, 1$, $\mathrm{Aut}(0) = C_2$, $\mathrm{Aut}(1) = 1$, a single arrow $a : 0 \to 1$ with $ag = a$, and no reverse arrow. It has four arrows in total and is checked in `G_checks.py`. Its realization is the cone on BC_2, with the cone point closed. One-object EI categories are groupoids; every finite acyclic case has already been settled. Thus this is a smallest mixed candidate by those two size measures, not an asserted counterexample. We make no claim that this open clause is unknown in the literature.

B.2 The endomorphism obstruction and marked replacements

B.6 (Proved). Presheaf categories remember idempotent completion. *For small ordinary categories S, T, an equivalence $\mathcal{P}(S) \simeq \mathcal{P}(T)$ exists if and only if their idempotent completions* $\mathrm{Kar}(S)$ *and* $\mathrm{Kar}(T)$ *are equivalent. More generally an equivalence $\mathcal{P}(S) \simeq \mathrm{Fun}(\mathcal{E}^{\mathrm{op}}, \mathcal{S})$ for a small ∞-category $\mathcal{E}$ identifies their idempotent completions. If every endomorphism of $\mathcal{E}$ is an equivalence, such an equivalence forces every endomorphism of S to be invertible.*

Proof. Call an object A of a presentable ∞-category atomic if $\mathrm{Map}(A, -)$ preserves all small colimits. Every representable presheaf is atomic, because its mapping-space functor is evaluation. Every presheaf is the colimit of its diagram of representables indexed by its category of elements. If A is atomic, apply $\mathrm{Map}(A, -)$ to that colimit presentation of A. The component of id_A is represented at some object of the indexing category: connected components of a colimit of spaces are generated by points of its terms. Thus there are maps $A \to h_s \to A$ with composite homotopic to id_A. Conversely a retract of a representable is atomic, since preservation of a colimit is closed under retracts. The full subcategory of atomic objects is therefore exactly the idempotent completion of the represented category. An equivalence of presheaf categories preserves this intrinsic subcategory. Conversely functors to an idempotent-complete category extend uniquely across idempotent completion, so equivalent completions give equivalent presheaf categories.

Suppose every endomorphism of $\mathcal{E}$ is an equivalence. Every idempotent in it is then an equivalence and splits; equivalently its image in the homotopy category is an invertible idempotent and therefore an identity, and the coherent splitting has a contractible space of choices. Thus $\mathcal{E}$ is idempotent complete. The fully faithful embedding $S \to \mathrm{Kar}(S)$, followed by the equivalence to $\mathcal{E}$, reflects equivalences. Its image endomorphisms must be equivalences, so those of S are invertible. This is the presheaf Morita argument; no claim about the bare ordinary homotopy category is used. □

B.7 (Refuted). All small categories as usual stratified exit models. *If S has a noninvertible endomorphism, no small layered exit ∞-category can have presheaf ∞-category equivalent to $\mathcal{P}(S)$. In particular the proposed geometric dictionary fails for any exodromic space stratified over a poset with those coefficients. The smallest example is the one-object monoid $M = \{1, e\}$, $e^2 = e \neq 1$; its classifying space is nevertheless contractible.*

Proof. An endomorphism in a layered ∞-category maps to an identity in its stratifying poset and is therefore an equivalence by conservativity. Apply B.6, taking the opposite if necessary to express the covariant exit convention as a presheaf convention. This proves the obstruction even to an abstract equivalence of the enhanced coefficient categories, not just to a particular map from S to a poset.

For the finite example all compositions are multiplication in $\{1, e\}$ with e absorbing, so associativity is immediate. The functor from M through the terminal category back to M sends both arrows to 1. The component e is a natural transformation from the identity functor to this constant functor, since $em = e = 1e$ for both m. Consequently BM contracts. The endomorphism e of the representable presheaf is still noninvertible by Yoneda; B.6 prevents its disappearance under any equivalence of presheaf categories. Its Karoubi completion has two objects $1, e$, endomorphism sets $\{1, e\}$ and $\{e\}$, and one arrow in each off-diagonal hom-set. The program verifies these sets and their composition. A category with fewer than one object has no endomorphisms, and a one-arrow category has only its identity, proving minimality in numbers of objects and arrows. □

B.8 (Known). What a subdivision can retain. *For every small category S there is a small poset Q and a functor $v : NQ \to NS$ which is an ∞-categorical localization at the arrows W_v sent to equivalences. Consequently, for every ∞-category $\mathcal{C}$,*

$$\mathrm{Fun}(S^{\mathrm{op}}, \mathcal{C}) \simeq \{F \in \mathrm{Fun}(Q^{\mathrm{op}}, \mathcal{C}) : F(w) \text{ is an equivalence for all } w \in W_v\}.$$

This is a marked-poset presentation, not an equivalence with all coefficients on the unmarked poset. The unmarked classifying space retains only BS.

Proof. Apply Kerodon (2026a, Theorem 6.3.6.1) to NS. Its universally localizing map is in particular a localization; maps between nerves of ordinary categories are functors. Its universal property gives the displayed fully faithful restriction and its stated essential image, after taking opposites. Localizing further at every arrow gives $BQ \simeq BS$, since NS is already obtained by localizing a subset. In the presence of a noninvertible endomorphism, B.7 proves that the markings cannot be interpreted as an ordinary conservative stratification whose unrestricted constructible category is the desired one. They are extra coefficient equations. The category of simplices and its last-vertex map give another explicit localization presentation, by Corollary 6.3.6.5 of the same source. A weak equivalence from an arbitrary subdivision alone supplies no such coefficient comparison. □

B.3 Fine and coarse information

B.9 (Proved). The two invariants and their strict difference. *The enhanced fine invariant is the Morita class* $\mathrm{Kar}(S)$*, equivalently $\mathcal{P}(S)$. Its locally constant part is* $\mathrm{Fun}(BS, \mathcal{S})$ *and depends only on the groupoid completion BS. Morita equivalence implies equivalence of classifying spaces, but the converse is false. Neither assertion identifies a bare equivalence of ordinary categories* $\mathrm{HOT}(S)$ *with enhanced Morita equivalence.*

Proof. B.6 proves the fine assertion, and A.9 proves the locally constant one. For completeness, the inclusion $J : S \to \mathrm{Kar}(S)$ induces a weak homotopy equivalence on nerves. An object of the comma category $((s, e)\backslash J)$ is a map $a : s \to t$ satisfying $ae = a$. The distinguished object is $e : (s, e) \to J(s)$. For each a its underlying map $a : s \to t$ defines a morphism from that distinguished object to a. These arrows constitute a natural transformation from its constant functor to the identity, because a comma morphism b satisfies $ba = a'$. The comma category is contractible, even though the distinguished object need not be initial. Quillen's Theorem A proves BJ is a weak equivalence. Thus equivalent Karoubi completions have equivalent classifying spaces.

Let $S = e$ and $T = [1]$. Both classifying spaces are contractible, but $\mathcal{P}(e)$ has two subterminal objects and $\mathcal{P}([1])$ has three. A subterminal presheaf has each value either empty or a point, and its nonempty objects form a sieve. The sieves of e are $\varnothing, e$; those of $[1]$ are $\varnothing, \{0\}, \{0, 1\}$. An equivalence preserves the terminal object and its subobjects, so the presheaf categories differ. The finite computation records these subobject counts. This refutes the proposed equality of the fine and coarse theories on the smallest nontrivial poset. □

B.10 (Proved). Precisely what reconstructs the coarse space. *For a small space X, the ∞-topos $\mathrm{Fun}(X, \mathcal{S})$ has shape the constant pro-space X. Its ordinary homotopy category together with its specified constant-diagram functor $\Delta_X : \mathrm{Ho}\,\mathcal{S} \to \mathrm{Ho}\,\mathrm{Fun}(X, \mathcal{S})$ determines X in $\mathrm{Ho}\,\mathcal{S}$. Set-valued local systems with their full family of evaluation functors determine $\Pi_1 X$. For connected pointed X and a nonzero commutative coefficient ring spectrum R, the enhanced R-linear category $\mathrm{Fun}(X, \mathrm{Mod}_R)$ together with evaluation at its point determines the monad, and hence the E_1-algebra, $R[\Omega X]$.*

Proof. The terminal object is intrinsic and

$$\mathrm{Map}_{\mathrm{Fun}(X,\mathcal{S})}(1, \Delta_X K) \simeq \mathrm{Map}_{\mathcal{S}}(X, K).$$

Indeed the left side is the limit of the constant K over the groupoid X, which is its space of maps from X. This is the defining shape formula. On connected components it gives $\mathrm{Hom}(1, \Delta_X K) = [X, K]$ naturally in K; ordinary Yoneda in the homotopy category proves reconstruction with the specified Δ_X.

For sets, local systems are functors on $\Pi_1 X$. Evaluation at x is represented by the covariant representable $\mathrm{Hom}_{\Pi_1 X}(x, -)$. Yoneda identifies transformations between the evaluation functors with arrows in the fundamental groupoid, with the composition given by composition of those transformations. Remembering the whole family therefore reconstructs that groupoid. Restriction to finite-valued functors remembers its finite-action invariant instead; this is the finite-quotient qualification and is not a claim to recover an arbitrary discrete group from its finite actions.

For the linear assertion, evaluation U_x is conservative because all objects of a connected groupoid are equivalent, and it preserves colimits since they are pointwise. Its left adjoint L_x satisfies

$$(L_x M)(y) = R[\mathrm{Map}_X(x, y)] \otimes_R M.$$

The monad $U_x L_x$ is tensor product with $R[\Omega_x X]$; multiplication is induced by composition of paths. The monadicity theorem (Lurie, 2017, Theorem 4.7.3.5) applies because U_x is conservative and preserves geometric realizations. Thus remembering U_x remembers this particular monad and its algebra, while forgetting it leaves only the equivalence class of the module category. This verifies the enhanced-evaluation qualification in GALOIS F.1–F.3. □

B.11 (Proved). The specified map detected by each coefficient theory. *Let $f : X \to Y$ be a map of small spaces of CW homotopy type. The class of maps for which the specified restriction $f^* : \mathrm{Fun}(Y, \mathcal{C}) \to \mathrm{Fun}(X, \mathcal{C})$ is an equivalence is as follows.*

$\mathcal{C}$	*necessary and sufficient condition on f*
$\mathcal{S}_{\leq r},\ r \geq 0$	*$\tau_{\leq r+1} f$ is an equivalence*
$\mathcal{S}$	*f is a weak homotopy equivalence*
$\mathrm{Mod}_{H\mathbb{Z}}$	*f is a weak homotopy equivalence*
Sp	*f is a weak homotopy equivalence*
$\mathrm{Mod}_{H\mathbb{Q}}$	*$\pi_0 f, \pi_1 f$ are isomorphisms and $\pi_k f \otimes \mathbb{Q}$ is an isomorphism for $k \geq 2$*
0	*every map.*

Here the homotopy-group conditions are imposed at every basepoint and $\pi_0 f$ means a bijection. The same tests apply to equivalence of the ordinary homotopy categories under this specified restriction. In particular set-valued locally constant coefficients detect W_1, not W_0.

Proof. We first explain how equivalence on homotopy categories detects equivalence of the enhanced restriction. For A, B in the target coefficient category of diagrams and any small space K, cotensoring gives

$$\mathrm{Hom}_{\mathrm{Ho}}(A, B^K) = [K, \mathrm{Map}(A, B)].$$

Restriction preserves these pointwise cotensors. Full faithfulness on homotopy categories consequently gives a bijection on the displayed sets of homotopy classes for every K. Yoneda in $\mathrm{Ho}\,\mathcal{S}$ shows that the induced map of mapping spaces is a weak equivalence. Essential surjectivity is unchanged by passing to the homotopy category.

For truncated spaces, first restrict an equivalence to its intrinsically 0-truncated objects. This gives equivalence on set-valued local systems. The unit on the representable based at x identifies the path sets in $\Pi_1 X$ with those in $\Pi_1 Y$. A missing component of Y would allow two different local systems there with the same restriction; two components of X sent to one component would contradict that unit, whose value on the other component is empty on one side and nonempty on the other. Hence $\pi_0 f$ is bijective and $\pi_1 f$ is an isomorphism. The left adjoint to evaluation at x in r-types sends a point to $y \mapsto \tau_{\leq r} \mathrm{Map}_X(x, y)$. Since $f_!$ sends this free object to the corresponding free object at $f(x)$, its unit evaluated at x gives

$$\tau_{\leq r} \Omega_x X \longrightarrow \tau_{\leq r} \Omega_{f(x)} Y.$$

It is an equivalence, proving all required homotopy-group isomorphisms. Conversely every level of the complete Segal-space nerve of $\mathcal{S}_{\leq r}$ is an $(r+1)$-type: its components are objects or finite diagrams with equivalences, whose loop spaces are r-truncated mapping spaces. Maps into each such level depend only on $\tau_{\leq r+1} X$. Thus $\tau_{\leq r+1} f$ an equivalence makes the functor categories equivalent. The same free-object argument without truncation proves the row for spaces.

For a nonzero linear coefficient ring, the same free objects rule out missing or merged components, using zero versus the nonzero free module. On a connected component, B.10 identifies the specified restriction with restriction of modules along

$$R[\Omega_x X] \longrightarrow R[\Omega_{f(x)} Y]. \tag{B.11}$$

If restriction is an equivalence, its extension-of-scalars adjoint has invertible unit on the free rank-one module; this unit is (B.11) as a map of underlying modules. Conversely an equivalence of these E_1-algebras induces equivalence on modules.

For $R = H\mathbb{Z}$, (B.11) is an integral homology equivalence of loop spaces. In degree zero it is the group-ring map induced by the actual map of fundamental groups. A map between their group bases can be an isomorphism on free abelian groups only if it is bijective, so $\pi_1 f$ is an isomorphism. Restricting to the identity components gives an integral homology equivalence of the loop spaces of the universal covers. The simplicial bar construction preserves this equivalence: its levelwise product maps are homology equivalences by the integral Künneth theorem, and its first-quadrant homology spectral sequence converges in each total degree. Their deloopings, the simply connected universal covers, are therefore homology equivalent. The relative Hurewicz theorem, applied to a first possible nonzero relative homotopy group, makes a homology equivalence between simply connected CW types a weak homotopy equivalence. This proves the integral row.

For spectra, an equivalence of (B.11) implies the integral-chain equivalence after smashing with $H\mathbb{Z}$, so the preceding argument applies. The converse in both rows follows from homotopy invariance. For $H\mathbb{Q}$, the loop-chain characterization of Rivera, Wierstra, and Zeinalian (2021, Theorem 3) says exactly that (B.11) is an equivalence if and only if the displayed π_1-rational conditions hold. Its hypotheses allow arbitrary connected pointed spaces; applying it componentwise gives our formula. Finally the zero coefficient category has one object and one map on every base, proving the last row. This proves the coefficient table from its map-level tests and does not infer it from an abstract Morita equivalence of loop algebras. □

B.12 (Open). Reconstruction from bare ordinary coefficient categories. *Without a specified enhancement, constant-diagram functor, or evaluation functor, this audit does not prove that an abstract equivalence of ordinary categories* $\mathrm{HOT}^{\mathrm{lc}}(S)$ *and* $\mathrm{HOT}^{\mathrm{lc}}(T)$ *forces* $BS \simeq BT$*, or that an abstract equivalence* $\mathrm{HOT}(S) \simeq \mathrm{HOT}(T)$ *forces* $\mathrm{Kar}(S) \simeq \mathrm{Kar}(T)$*. Likewise it does not reconstruct the particular loop algebra from a bare ordinary category of linear local systems.*

The attempted proofs stop at missing data. B.6 characterizes atoms using mapping spaces and all coherent colimits. B.10 uses the specified Δ_X or U_x. B.11 uses a particular restriction functor which preserves cotensors, not an arbitrary equivalence of ordinary categories. None of these structures has been reconstructed from the bare category here. The exact smallest class not settled by these arguments consists already of connected bases with unspecified constant functor; we do not claim a minimal number of objects or a counterexample. This retains, without strengthening, the open qualification of GALOIS F.8.

C Gluing and links

C.1 (Proved). The categorical gluing datum is a bimodule. *Let $A \subset X$ be a full sieve in a small category, let $B = X \setminus A$ be its full complementary cosieve, and write $j : A \hookrightarrow X$ and $i : B \hookrightarrow X$. The datum of X with these specified inclusions is equivalent to the datum of A, B and*

$$M : A^{\mathrm{op}} \times B \longrightarrow \mathbf{Set}, \qquad M(a,b) = \mathrm{Hom}_X(a,b).$$

Its category of crossing arrows E has objects (a,b,m), $m \in M(a,b)$, and arrows $(u,v) : (a,b,m) \to (a',b',m')$ satisfying $u : a \to a'$, $v : b \to b'$ and $vm = m'u$. Its source and target are functors $s : E \to A$ and $t : E \to B$.

Proof. There is no arrow from B to A, since A is a sieve. All remaining hom sets are those of A, those of B, and the values of M. The functoriality of M gives precomposition by arrows of A and postcomposition by arrows of B. Its interchange identity is precisely associativity for a composite containing one crossing arrow. A composite can contain at most one crossing

arrow, so there are no other associativity conditions. Conversely these operations recover the stated functor. On morphisms one uses functors on A, B and a natural transformation of their crossing bimodules. Thus the reconstruction is an equivalence of categories of specified gluing data, with the evident canonical isomorphism back to the original X. This verifies XII ms. 59–60; the printed claim of an isomorphism of categories requires chosen object sets, as the editorial correction already notes. No localization or choice of paths occurs here. □

C.2 (Known). The crossing category is the homotopy link, with the correct complementary open subset. *Form $K = N(X^{\mathrm{op}}) \to \Delta[1]$ by assigning B to 0 and A to 1. Give $Z = |K|$ the last-vertex stratification over $\{0 < 1\}$ with its upper Alexandrov topology. Then its closed stratum is $Z_0 = B(B^{\mathrm{op}})$ and its open stratum is $Z_1 = Z \setminus Z_0$. There is a natural weak equivalence*

$$B(E^{\mathrm{op}}) \longrightarrow \mathrm{holink}(Z_0, Z_1),$$

where a homotopy-link path starts in Z_0 and lies in Z_1 at every positive time. Under the canonical opposite-nerve homeomorphisms, Z is BX, Z_0 is BB, and Z_1 deformation retracts onto BA. The comparison is compatible with both endpoint maps, followed at the open end by this deformation retraction.

Proof. An n-simplex of $\mathrm{Map}_{\Delta[1]}(\Delta[1], K)$ is a functor $[n] \times [1] \to X^{\mathrm{op}}$ which has its first row in B^{op} and its second in A^{op}. Reversing its arrows identifies it with an n-simplex of $N(E^{\mathrm{op}})$. This is an equality of simplicial sets, including faces and degeneracies. The realization map to topological homotopy links is the weak equivalence of Douteau and Waas (2022, Theorem 6.1), applied to this particular stratified simplicial set. That theorem applies to arbitrary stratified simplicial sets; no EI, finiteness, or Quinn-fibration condition has been added implicitly.

For the claimed retraction, represent a point by barycentric coordinates on a simplex of $N(X^{\mathrm{op}})$. A point belongs to Z_1 exactly when the total weight w of its A vertices is positive. Multiply every B coordinate by $1 - r$ and every A coordinate by $(1 - r) + r/w$, for $0 \le r \le 1$. The sum remains one, the coordinates remain nonnegative, and the formula commutes with faces and degeneracies. It defines a strong deformation retraction of Z_1 onto $B(A^{\mathrm{op}})$. The realization comparison in the cited theorem is obtained by evaluating the prism $|\Delta[n] \times \Delta[1]|$; its endpoints are therefore the source and target evaluations, not an unspecified abstract weak equivalence. Opposite-nerve homeomorphisms reverse the order of barycentric coordinates and identify these maps with Bs and Bt.

For uncountable categories Z need not be metrizable, so it need not be a homotopically stratified set in Quinn's original sense. Quinn's tameness and homotopy-link fibration hypotheses remain genuine additional hypotheses (Quinn, 1988, §§2.1,3.1). The theorem just used is the simplicial/stratified-space form of the comparison and avoids that unnecessary restriction. In particular BA and BB are not being asserted to be literal complementary subsets of BX: both are subcomplexes, and the complementary open set is $BX \setminus BB$. □

C.3 (Known). The specified categorical and homotopy-link gluing maps agree. *For the data of C.1, the comparison determined by the natural transformation $js \Rightarrow$ it is a weak homotopy equivalence*

$$\mathrm{hocolim}(BA \xleftarrow{Bs} BE \xrightarrow{Bt} BB) \longrightarrow BX.$$

Via C.2 it is the homotopy-link double-mapping-cylinder comparison. This includes empty A, empty B, empty E, and arbitrarily many connected components.

Verification of the supplied proof and comparison. We use the supplied POLES G.19 proof rather than give a third proof. Its category P has the three fibres A, E, B, arrows $a' \to (a \to b)$ indexed by $a' \to a$, and arrows $b' \to (a \to b)$ indexed by $b' \to b$. The displayed functor $P \to X$ sends a crossing-arrow object to its target. For $x \in A$, its comma category is A/x. For $x \in B$, the endofunctor printed there sends $a \to x$ to the corresponding crossing arrow, sends $(a \to b \to x)$ to its composite, and sends an object of B/x to id_x. The two printed natural transformations are well typed: on a crossing object their component is the square $(\mathrm{id}_a, b \to x)$, and on the other fibres it is the indicated structural arrow. The squares commute by the definition of E; the constant functor involved takes its value at the B object (x, id_x). Thus every comma category is contracted exactly as claimed. The fibred Grothendieck-construction comparison identifies BP with the stated homotopy colimit and identifies its map to BX with the source-to-target homotopy. These checks validate the application of Theorem A in POLES G.19 (GPT-6 Astra (OpenAI), 2026a).

The same source-to-target prism occurs in C.2. After reversing its interval coordinate to start at BB, and applying the deformation retraction at its open endpoint, it is the holink evaluation prism. Hence the comparison is the same coherent cocone. This derives the weak homotopy-link gluing assertion for these nerves from the audited categorical proof. It does not assert Quinn's stronger complement-preserving homotopy equivalence without his tameness hypotheses. If one of the pieces is empty, the corresponding crossing category is empty and the homotopy colimit reduces to the remaining piece. If E alone is empty, $X = A \amalg B$ and it reduces to a disjoint union. □

C.4 (Refuted). Unrestricted paths cannot replace the crossing link. *Replacing the crossing category in C.3 by all paths between its endpoint subspaces changes the answer, already for a category with two objects and four arrows.*

Proof. Let X have objects a, b, identity arrows and exactly two further arrows $r, s : a \to b$. Composition with identities is the only composition. Take $A = \{a\}$ and $B = \{b\}$. Then E is the discrete two-object category and BX is the circle obtained from two edges with their endpoints identified in pairs. Consequently the unrestricted path space from a to b is homotopy equivalent to ΩS^1 and has components indexed by $\mathbb{Z}$, each contractible. In contrast BE has exactly two components. The double mapping cylinder of two points under $\mathbb{Z}$ has first integral homology freely generated by countably many differences of edges, whereas the double mapping cylinder under $\{r, s\}$ has first homology $\mathbb{Z}$. Thus even the resulting unstratified gluing type changes. The category and its nerve boundary matrix are checked in `code/CD_checks.py`; its output records Betti numbers $(1, 1)$. The distinction is between paths constrained to leave the closed stratum immediately and paths allowed to return to it and acquire arbitrary winding. □

C.5 (Proved). Finite filtrations require the actual crossing categories and their flag data. *Let $\varnothing = A_0 \subset A_1 \subset \cdots \subset A_r = X$ be a finite filtration by full sieves, and put $B_k = A_k \setminus A_{k-1}$. At stage k let E_k be the category of arrows in A_k from A_{k-1} to B_k. Then*

$$BA_k \simeq \mathrm{hocolim}(BA_{k-1} \leftarrow BE_k \to BB_k)$$

by the specified comparisons of C.3. All generalized links are given by the nerves of the categories of flagged strings of arrows: for $i_0 < \cdots < i_m$, their objects are $b_0 \to \cdots \to b_m$ with $b_j \in B_{i_j}$ and their morphisms are commuting ladders. These categories, their restriction functors, and their composition functors retain the iterated gluing information.

Proof. Each A_{k-1} is a sieve in A_k, so C.3 applies successively. This is an induction using actual categories and actual functors. There is no appeal to an isomorphism of pro-objects or to a limit of homotopy categories. The level function $X \to \{1 < \cdots < r\}$ is a functor; its opposite stratifies $N(X^{\mathrm{op}})$ over the oppositely ordered level set. Applying the simplicial calculation of C.2 to the reversed flag instead of [1] identifies its simplicial mapping space with the nerve of the opposite of the specified ladder category. Douteau–Waas Theorem 6.1 again supplies its geometric interpretation. Restriction of strings supplies the face maps, and ordinary composition in X supplies the remaining flag maps; all their identities hold before realization.

One must not replace these data by pairwise homotopy types without their coherence. Even the boundary of a triangle and the triangle, stratified by their ordered vertices, have equivalent individual strata and pairwise holinks but different three-stratum links: the full-flag holink is empty for the boundary and nonempty for the triangle (Douteau & Waas, 2022, Example 2.30). Their underlying spaces also have different first homology. The accompanying program checks the boundary matrices. When every layer is a groupoid, the ladder categories are groupoids and their endpoint maps are isofibrations (lift an endpoint isomorphism by whiskering); their strict Segal fibre products then are homotopy fibre products. This is the spatial décollage situation. For arbitrary non-groupoidal layers one retains the flagged categories themselves; groupoidification has not been claimed to preserve arbitrary strict fibre products. □

C.6 (Proved). Enhanced coefficients glue by a lax pullback. *Let $\mathcal{C}$ be a presentable ∞-category, and put $\mathcal{F}_C = \mathrm{Fun}(C^{\mathrm{op}}, \mathcal{C})$ for a small category C. Define*

$$T = i^* j_* : \mathcal{F}_A \longrightarrow \mathcal{F}_B.$$

Then $\mathcal{F}_X$ is equivalent to the ∞-category of triples $(F_A, F_B, \theta : F_B \to TF_A)$. There is a canonical equivalence $T \simeq t_* s^*$, so the gluing arrow may equivalently be specified as a homotopy-coherent natural transformation $t^*F_B \to s^*F_A$ in $\mathcal{F}_E$. At $b \in B$,

$$(TF_A)(b) \simeq \lim_{(A/b)^{\mathrm{op}}} F_A.$$

The limit is an ∞-categorical limit; neither it nor the lax pullback may be replaced by its ordinary counterpart in homotopy categories.

Proof. An object $F \in \mathcal{F}_X$ has the unit $F \to j_* j^* F$, whose restriction to B gives θ. Conversely define, from the given coherent arrow, an object of $\mathcal{F}_X$ by

$$F = j_* F_A \underset{i_* TF_A}{\times} i_* F_B.$$

The map from the first factor is the unit of $i^* \dashv i_*$ and the map from the second is $i_*\theta$. Restriction to A gives F_A, since i_* has terminal values there. Restriction to B gives $TF_A \times_{TF_A} F_B \simeq F_B$. Applied to an original F, its two units therefore give a comparison to this pullback which is an equivalence at every object, hence an equivalence of functors. Applied to a triple, the same restrictions recover its specified arrow. The constructions use functorial limits and units, so these are inverse equivalences of ∞-categories, including mapping spaces.

To identify T, consider $r : E/b \to A/b$, $(a \xrightarrow{m} b', \beta : b' \to b) \mapsto (a, \beta m)$. It has a fully faithful right adjoint $k : A/b \to E/b$, $(a, n : a \to b) \mapsto (n, \mathrm{id}_b)$. The adjunction is the equality between a square with target component β and its source component $a \to a'$. Thus k is homotopy final, k^{op} is homotopy initial, and the pointwise right-Kan formula identifies $t_* s^* F_A$ with the displayed limit over $(A/b)^{\mathrm{op}}$. Adjunction $t^* \dashv t_*$ now identifies the two spaces of gluing arrows, naturally in both coefficient objects. The ordinary set-valued case is the gluing of

topoi reproduced at XII ms. 62–63 from SGA 4, Exposé IV, Theorem 9.5.4 and Exercise 9.5.10 (Artin, Grothendieck, & Verdier, 1972). The argument above proves the enhanced version directly, rather than inferring it from ordinary sheaf gluing. □

C.7 (Refuted). The ordinary comma category of homotopy categories loses gluing morphisms. *Even for $X = [1]$ and space-valued coefficients, the homotopy category of C.6 is not reconstructed by replacing each ∞-category in its lax pullback by its homotopy category and taking an ordinary comma category.*

Proof. View two objects of $\mathrm{Fun}([1]^{\mathrm{op}}, \mathcal{S})$ as the arrows $f : S^0 \to *$ and $g : * \to S^1$. Their mapping space is

$$\mathrm{Map}(S^0, *) \underset{\mathrm{Map}(S^0, S^1)}{\overset{h}{\times}} \mathrm{Map}(*, S^1) \simeq * \underset{S^1 \times S^1}{\overset{h}{\times}} S^1,$$

where the last map is the diagonal. Its connected components are the cokernel of $\mathbb{Z} \xrightarrow{(1,1)} \mathbb{Z}^2$, hence form $\mathbb{Z}$. In the ordinary arrow category of $\mathrm{Ho}\,\mathcal{S}$, both vertical maps in such a square have a unique homotopy class and their composites agree, so there is exactly one morphism from f to g. Thus the comparison is not faithful. The integer boundary-rank check is included in the program. For a bare derivator, the Kan extensions, support subderivators, and the pointed recollement proved in Part D remain meaningful; the false ordinary comma formula is not a substitute for an enhancement. □

C.8 (Proved). Locally constant enhanced coefficients glue by an ordinary ∞-categorical pullback. *For any complete and cocomplete ∞-category $\mathcal{C}$, the locally constant part is*

$$\mathrm{Fun}(BX, \mathcal{C}) \simeq \mathrm{Fun}(BA, \mathcal{C}) \underset{\mathrm{Fun}(BE, \mathcal{C})}{\times} \mathrm{Fun}(BB, \mathcal{C}),$$

where the pullback is in $\mathbf{Cat}_\infty$ and therefore uses an equivalence between the two restricted local systems. The same assertion holds for every stage of C.5.

Proof. Groupoidification identifies locally constant functors on X^{op} with functors on BX. C.3 is a pushout in spaces, and its image is a pushout in $\mathbf{Cat}_\infty$: the inclusion of spaces is left adjoint to the maximal-subgroupoid functor and hence preserves colimits. Functor categories turn a colimit in their first variable into a limit, giving the stated equivalence. More explicitly, a functor out of the double mapping cylinder consists of two functors and a natural equivalence along BE; all their higher compatibility is part of the mapping-space pullback. The same argument applied at each stage proves the finite-filtration assertion. Taking Ho after forming this pullback gives the locally constant homotopy category, but does not turn it into an ordinary categorical pullback. The abstract-derivator invariance theorems of the supplied GALOIS B.3 and B.5 do not change that distinction. □

C.9 (Proved). The exact weight test for a homotopically valid strict enriched gluing. *For a sieve $U \hookrightarrow X$ let $H_t(u) = \mathrm{Hom}_X(u, t)$, $t \in X \setminus U$. Regard ordinary categories as simplicial categories with discrete hom objects. Cobase change along this sieve preserves all Dwyer–Kan equivalences of simplicial categories under U if and only if each degreewise coend*

$$H_t \otimes_U - : \mathrm{Fun}(U, \mathbf{sSet}) \longrightarrow \mathbf{sSet}$$

preserves objectwise weak homotopy equivalences. This condition, discrete flatness, is stronger than the homotopical cofibration condition for classifying spaces.

Proof. In the pushout against $p : U \to C$, the new hom object from $c \in C$ to t is precisely $H_t \otimes_U C(c, p(-))$; all other hom objects are unchanged or empty. This follows by writing a crossing composite as one arrow in C, one arrow in X crossing the sieve, and imposing the usual balanced relations. Thus preservation by all weights gives full faithfulness up to weak equivalence after pushout. Essential surjectivity follows from that of the original Dwyer–Kan equivalence, because the complementary objects are unchanged.

Conversely, for any $L : U \to \mathbf{sSet}$ adjoin a new source c to U, with hom objects $C_L(c, u) = L(u)$ and no nonidentity arrows into c. An objectwise weak equivalence $L \to L'$ gives a Dwyer–Kan equivalence $C_L \to C_{L'}$. After pushout its hom map from c to t is the weight map under examination. Preservation of every Dwyer–Kan equivalence therefore forces the asserted condition. This independently checks the proof of TOTASPH F.14 (GPT-6 Astra (OpenAI), 2026b).

For the claimed implication to homotopical cofibrations we use the generalized version of Hackney, Ozornova, Riehl, and Rovelli (2024, Theorem 1.5, §4, p. 2180): for a discretely flat inclusion the comparison $NX \amalg_{NU} NC \to N(X \amalg_U C)$ is a categorical equivalence. Since $NU \to NX$ is a monomorphism, its source computes the homotopy pushout of spaces. Comparing this construction for a weak equivalence $C \to C'$ under U and using homotopy-pushout invariance proves the implication. C.11 proves that the converse fails. In particular a homotopy colimit in C.3 has not been exchanged for a colimit of weights without a hypothesis. □

C.10 (Proved). A universally harmless attachment can have a disconnected individual crossing category. *There is a three-object, seven-arrow category X and a one-object sieve U such that $U \to X$ remains a weak equivalence after every pushout, while U/t has two components for a complementary object t.*

Proof. Take objects a, t, s, nonidentity arrows $\alpha, \beta : a \to t$, $\gamma : t \to s$, and $\delta : a \to s$, with $\gamma\alpha = \gamma\beta = \delta$. Set $U = \{a\}$. These equations and identities specify all compositions. For a functor $U \to C$ selecting c, the pushout P has the old category C, the two new objects t, s, hom sets $P(d, t) = C(d, c) \times \{\alpha, \beta\}$, $P(d, s) = C(d, c)$, and $P(t, s) = \{\gamma\}$. There are no reverse crossing arrows. The retraction $r : P \to C$ sends both new objects to c and forgets the two-arrow label. The endofunctor $F : P \to P$ fixes C and sends t, s to s. There are natural transformations $\mathrm{id}_P \Rightarrow F \Leftarrow jr$, with components γ, id_s for the first and the arrow corresponding to id_c for the second; both are identities on C. Naturality at α, β is exactly their equality after composition with γ. Thus Bj is a homotopy equivalence for every C and every attaching functor. Yet U/t is discrete with two objects. The program checks all compositions, associativity, the sieve condition, both transformations on the original category, and the crossing components. This verifies the universal argument in TOTASPH F.18, rather than inferring universality from finite testing. The source U is discrete, so its weights are coproducts of copies and satisfy C.9. □

C.11 (Proved). A homotopical cofibration need not be discretely flat. *There is a four-object, eleven-arrow full sieve $U \hookrightarrow X$ which is a couniversal weak equivalence but fails the weight test of C.9.*

Proof. Let $U = (0 \xrightarrow{a} 1)$. Adjoin t and arrows $\beta, \gamma : 1 \to t$ and $c : 0 \to t$, with $\beta a = \gamma a = c$, and then adjoin a terminal object s. There are no other arrows except identities and the unique arrows into s. Send t, s to 1 to obtain a retraction $r : X \to U$. The endofunctor fixing U and sending t, s to s gives $\mathrm{id}_X \Rightarrow F \Leftarrow ir$, relative to U. On pushing out along any $U \to C$, these functors and transformations extend by the identity on C: their component equations are unchanged balanced-composite equations. Hence each pushout inclusion is a categorical

homotopy equivalence. Consequently cobase change preserves weak equivalences under U by two-out-of-three.

For non-flatness take a simplicial circle K and its cone CK. On U define $L = (K \hookrightarrow CK)$ and $L' = (K \to *)$, with comparison identity at 0 and the contraction at 1. This is an objectwise weak equivalence. The crossing weight H_t is singleton at 0 and has two elements at 1, both restricting to the singleton. Directly from the coend relations,

$$H_t \otimes_U L = CK \amalg_K CK, \qquad H_t \otimes_U L' = *.$$

The first is the suspension of the circle, with integral $H_2 = \mathbb{Z}$; its map to the second is not a weak equivalence. The program checks the finite category, its relative transformations, and a simplicial two-cone model with Betti numbers $(1, 0, 1)$. This completes the proposed TOTASPH F.21 test. It also explains why "the strict gluing has the right classifying space" and "all crossing weights compute their derived values strictly" are different assertions. □

D Recollement and base change

D.1 (Known). Pointed derivators have the exceptional adjoints in the manuscript's variance. *Let $\mathbb{D}$ be a derivator satisfying Der 1–4 on all small categories, in presheaf variance, and suppose that $\mathbb{D}(e)$ has a zero object. For the sieve $j : A \hookrightarrow X$ and complementary cosieve $i : B \hookrightarrow X$ there are adjunction chains*

$$j^? \dashv j_! \dashv j^* \dashv j_*, \qquad i_! \dashv i^* \dashv i_* \dashv i^!.$$

All four Kan extensions $j_!, j_*, i_!, i_*$ are fully faithful. The essential image of i_* consists of objects zero on A, and the essential image of $j_!$ consists of objects zero on B. These assertions hold at every additional diagram shape and do not require the strongness axiom Der 5.

Proof. The pointwise Kan-extension formulas and full faithfulness of the inclusions show that restriction of either Kan extension to its source is the identity. Outside B, the comma category computing i_* is empty; hence i_* has terminal, therefore zero, values there. The unit $F \to i_* i^* F$ is an isomorphism precisely when its values outside B are isomorphisms to zero, by Der 2. The corresponding argument for $j_!$ uses initial values outside A. This proves the essential-image descriptions and full faithfulness without a diagram-lifting axiom.

For existence of the outer adjoints, apply Groth (2013, Lemma 3.7 and Corollary 3.8) to the opposite-indexing derivator $I \mapsto \mathbb{D}(I^{\mathrm{op}})$. To make the variance and the construction explicit, the proof there uses the full subcategory of $K \times [1]$ on $(k, 0)$ and $(u(j), 1)$ for a sieve $u : J \hookrightarrow K$. Its projection q to K and its inclusion s of $K \times \{0\}$ give the reflector $q_! s_*$ onto diagrams zero on J; the dual cylinder gives the coreflector. The pointwise comma categories over $u(j)$ are $(J/j) \times [1]$ and have the indicated terminal objects, so the reflector vanishes exactly where required. Transporting this construction to presheaf variance gives a coreflector onto $\operatorname{im} i_*$ and a reflector onto $\operatorname{im} j_!$. Composing with the inverse restriction equivalences gives $i^!$ and $j^?$. The proof uses Kan extensions and pointwise conservativity only and applies unchanged after shifting by any small category. This is precisely the additional adjunction statement of I ms. 1–3, rather than an assertion that an arbitrary Kan extension has further adjoints. □

D.2 (Known). The pointed sequences and the stable recollement. *Under D.1 there are canonical coherent fibre and cofiber constructions*

$$i_*i^!F \simeq \mathrm{fib}(F \to j_*j^*F), \qquad j_!j^?F \simeq \mathrm{cofib}(i_!i^*F \to F),$$

and the canonical square with sides $j_!j^*F \to F \to i_*i^*F$ and its fourth vertex zero is both homotopy cartesian and homotopy cocartesian. If $\mathbb{D}$ is stable, these give the coherent exact sequences

$$j_!j^*F \longrightarrow F \longrightarrow i_*i^*F, \qquad i_*i^!F \longrightarrow F \longrightarrow j_*j^*F.$$

If in addition $\mathbb{D}$ is strong, their canonical triangles give the usual triangulated recollement $(j_!, j^*, j_*; i^*, i_*, i^!)$ at every shape. Pointedness alone does not assert stability or a triangulated structure.

Proof. The support coreflection in D.1 is represented by the fibre of the unit to j_*j^*; its restriction to A is the fibre of an identity, hence zero. The universal support adjunction identifies it with $i_*i^!$. Dually the support reflection is the cofiber of the counit from $i_!i^*$ and identifies with $j_!j^?$. These are coherent derivator fibres and cofibers. They are not ordinary fibres or cokernels in $\mathbb{D}(X)$.

The special adjunction maps are themselves obtained coherently by the mapping-cylinder Kan extensions used in D.1. We can construct the square without a diagram-lifting axiom. In $X \times [1] \times [1]$, with coordinates (x, t, s), let S be the sieve defined by $t = 0$ or $x \in A$, and let J be the cosieve defined by $s = 1$ or $x \in B$. Restrict the pullback of F to $S \cap J$, extend by zero along the sieve $S \cap J \hookrightarrow J$, and then extend by zero along the cosieve $J \hookrightarrow X \times [1]^2$. In presheaf variance the resulting square, with vertices $(1,1), (0,1), (1,0), (0,0)$, is

$$\begin{array}{ccc} j_!j^*F & \longrightarrow & F \\ \downarrow & & \downarrow \\ 0 & \longrightarrow & i_*i^*F. \end{array}$$

It restricts at $a \in A$ to a square with an identity $F_a \to F_a$ and two zero vertices, and at $b \in B$ to one with the other identity $F_b \to F_b$. Both are bicartesian. Pointwise detection of homotopy cartesian and cocartesian squares proves the assertion in the shifted derivator. This reasoning establishes a particular coherent square and does not invoke the false assertion that every incoherent square lifts uniquely.

Stability identifies fibres and cofibers with the expected suspension shift. Under Der 5, Groth's canonical triangulation theorem and Example 4.25 therefore give the recollement (Groth, 2013, Theorem 4.16, §4.3). Full faithfulness and the kernel-image equality are D.1, while the two triangles are the displayed sequences followed by their suspension connecting maps. All these six functors are exact in the stable sense. Their coherent Mayer–Vietoris sequences are also covered by Groth, Ponto, and Shulman (2014, Theorem 6.1). The manuscript's ordinary abelian short exact sequence in I ms. 2 is its represented abelian specialization; it is not a short exact sequence of spaces. In particular we retain the editors' pointedness correction before its square (2). □

D.3 (Proved). Exact finite-poset tests for properness and smoothness. *Let $f : P \to Q$ be a monotone map of small posets. In the manuscript's W_∞ terminology, f is proper if and only if for every $p \in P$ and every $q \geq f(p)$ the poset*

$$L_f^+(p,q) = \{p' \in P : p \leq p',\ f(p') = q\}$$

has weakly contractible nerve. It is smooth if and only if for every $p \in P$ and every $q \leq f(p)$ the poset

$$L_f^-(p,q) = \{p' \in P : p' \leq p,\ f(p') = q\}$$

has weakly contractible nerve. These tests apply to finite, locally finite and arbitrary small posets. Empty lifting posets fail the test. No condition is imposed for a pair (p, q) for which the displayed base inequality does not hold.

Proof. For a functor $f : X \to Y$ and an arrow $\alpha : f(x) \to y$, the outgoing lift category has objects $v : x \to x'$ with $f(x') = y$ and $f(v) = \alpha$, and morphisms $w : x' \to x''$ over 1_y satisfying $wv = v'$. This is the comma category $(x, \alpha)\backslash X_y$ for the canonical inclusion $X_y \to X/y$. The latter is homotopy final precisely when all these comma categories are weakly contractible. This is the properness condition of I ms. 53–54. Its dual is the incoming lift category with a fixed target upstairs, equivalently homotopy initiality of $X_y \to y\backslash X$. Maltsiniotis proves the equivalent universal base-change formulation (Maltsiniotis, 2005, Theorem 1.2 and its proper dual). For posets an arrow is uniquely determined by its endpoints, and the two lift categories become exactly the two full subposets in the statement. This proves both directions without replacing contractibility by connectedness. The empty category is not weakly contractible, so existence of a lift is a necessary part of either test. Identity-arrow tests have p itself as an initial or terminal object and are automatically satisfied. □

D.4 (Proved). Fibre formulas and universal base change. *Let*

$$\begin{array}{ccc} X' & \xrightarrow{g'} & X \\ {\scriptstyle f'}\downarrow & & \downarrow{\scriptstyle f} \\ Y' & \xrightarrow{g} & Y \end{array}$$

be a strict pullback of small categories. For every presheaf-variance derivator satisfying Der 1–4 the specified mate

$$g^* f_* \longrightarrow f'_* g'^*$$

is invertible if f is W_∞-proper or if g is W_∞-smooth. Its left-Kan counterpart

$$f'_! g'^* \longrightarrow g^* f_!$$

is invertible if f is smooth or g is proper. Properness of a fixed f is equivalent to the first comparison for every strict pullback and every space-valued coefficient; smoothness is equivalent to the dual assertion. For proper f and smooth f, respectively, the fibre comparisons are

$$(f_* F)_y \simeq \operatorname*{holim}_{X_y^{\mathrm{op}}} F|_{X_y}, \qquad (f_! F)_y \simeq \operatorname*{hocolim}_{X_y^{\mathrm{op}}} F|_{X_y}.$$

Proof. The ordinary pointwise index for f_* is $(X/y)^{\mathrm{op}}$; D.3 says that the opposite of $X_y \to X/y$ is homotopy initial. Thus restriction to the fibre computes the same homotopy limit. This cofinality is valid for all derivators, by Groth et al. (2014, Theorems 3.8 and 3.16). Similarly, for $f_!$ the index is $(y\backslash X)^{\mathrm{op}}$, and smoothness makes its fibre inclusion homotopy final. This proves the two displayed comparisons with their correct variances.

Under strict pullback the outgoing lift categories of f' identify with those of f at the prescribed base arrow: a lift of an arrow in Y' is a lift of its image in Y together with its already specified Y' component. Morphisms identify as well. Hence properness is preserved by strict pullback, as is smoothness by the dual argument. For proper f, the strict fibres of f over $g(y')$ and of f' over y' are isomorphic. The two fibre formulas therefore identify the right mate with an isomorphism at every y', and Der 2 proves right base change. Smooth f gives the left statement by the same argument. If g is smooth, transpose the square and apply this left statement to obtain $g'_! f'^* \to f^* g_!$; the transformation between their full right adjoints is exactly $g^* f_* \to f'_* g'^*$ and is invertible. The proper-g assertion is dual. This uses the

adjunction correspondence for whole composite adjoints, which preserves invertibility, rather than an unrestricted assertion about partial mates. Thus no additional topological assumption is hidden in the word "smooth."

For the converse to properness it suffices to test fibre squares. The adjoint mate of their right comparison is

$$k_! p^* \longrightarrow f^* y_!, \qquad k : X_y \hookrightarrow X, \quad p : X_y \to e.$$

Evaluate it on the terminal space. At $x \in X$ the target is the discrete representable $\mathrm{Hom}_Y(f(x), y)$; the source is the coproduct, indexed by $\alpha : f(x) \to y$, of the nerves of the outgoing lift categories. The map collapses each summand to its index. Its being an equivalence forces every such lift category to be weakly contractible. This is exactly D.3's general-category criterion. Passing to opposite indexing categories proves the smooth converse. The proof covers empty fibres and nonsurjectivity on components: an empty indexing family gives the corresponding initial or terminal coefficient, and an existing base arrow with no lift is detected by an empty summand over a nonempty representable. □

D.5 (Refuted). Topological properness and a stratified submersion do not characterize these functors. *For finite posets, every induced map $|Nf|$ is proper as a map of topological spaces. This does not imply categorical properness. Categorical properness does not imply that strict fibres are homotopy fibres. Categorical smoothness does not imply local product structure on realizations.*

Proof. A finite-poset nerve realizes to a compact Hausdorff polyhedron, so every continuous map between such realizations is proper. Let $f : e \to [1]$ select 0. It is smooth: its incoming lifting categories are singletons whenever a base arrow has target 0. It is not proper, since the arrow $0 \to 1$ has no outgoing lift. Its realization is the endpoint inclusion $\{0\} \hookrightarrow [0, 1]$, a proper map that has no local product description as a submersion onto a neighborhood of 0. This gives both the properness and the smooth-submersion separation, with the source and target completely specified.

The dual inclusion $h : e \to [1]$ selecting 1 is proper and not smooth. Its strict fibre at 0 is empty, while its homotopy fibre at 0 is contractible, because $[0, 1]$ is contractible and nonempty paths join 1 to 0. Its over-comma transition is $\varnothing \to e$, which is not a weak equivalence. Thus categorical properness guarantees the strict-fibre formula for contravariant right Kan extensions, not the Quillen-Theorem-B fibre comparison or locally constant transport without their further hypotheses. This independently verifies the particular proper-cosieve correction to row Ib-C003 in the supplied GALOIS D.9.

The first example also displays failure of right base change. Pull it back along the inclusion of 1. The strict pullback is empty. For a discrete two-point space M, the value of f_*M at 1 is M, since $e/1$ is terminal, whereas right Kan extension from the empty fibre has value $*$. The base-change map is $M \to *$ and is not invertible. The program checks the posets, both lift tests, and these cardinalities. There are no omitted endomorphisms or compositions in these posets. □

D.6 (Proved). Proper and smooth is strictly weaker than a strict Grothendieck bifibration. *Let P have four objects x, a, b, c and order relations $x < a < c$, $x < b < c$, with a, b incomparable. Let $f : P \to [1]$ send x to 0 and a, b, c to 1. Then f is proper and smooth, and is a Grothendieck fibration, but is not a Grothendieck opfibration. Its realization is not a locally trivial topological fibre bundle.*

Proof. The only nonidentity outgoing lifting test not confined to one fibre is $L_f^+(x, 1) = \{a, b < c\}$. It has terminal object c, hence contractible nerve. Every other outgoing test has its fixed

source as an initial object. The incoming lifting category for $0 \to 1$ at each of a, b, c is the singleton $\{x\}$; the remaining incoming tests have their fixed target as terminal object. D.3 proves properness and smoothness.

For a poset over this two-element base, a cartesian lift of $0 \to 1$ with fixed target exists exactly when the incoming lift poset has a greatest element. Indeed every base arrow into 0 is its identity, so the full cartesian factorization test is exactly the greatest-element test in that fibre. All the incoming lift posets just listed have the required greatest element x. Identity arrows of the base have identity lifts. Dually an opcartesian lift from x over $0 \to 1$ would be a least element of $\{a, b < c\}$. There is none, since a, b are incomparable minima. Thus f is not an opfibration. The program exhausts every lifting poset and checks these greatest/least-element tests over [1]. For bases with more composable arrows, greatest elements in exact incoming lift posets alone would only establish the weaker precartesian test; we make no such general cartesian criterion.

The realization is a cone on the interval-shaped realization of $\{a, b < c\}$, projected to $[0, 1]$ by its cone coordinate. The fibre at 0 is a point and each fibre at $t > 0$ is a nontrivial interval-shaped tree. These fibres are not homeomorphic, so there is no local trivialization at 0. They are all contractible, which also shows why weak contractibility of lift categories cannot supply a geometric bundle structure. Part E uses this distinction when auditing VIc-C008. □

D.7 (Proved). Inclusions, components, and empty sources. *For a full inclusion $P \hookrightarrow Q$ of posets, categorical properness is equivalent to P being a cosieve, and categorical smoothness is equivalent to P being a sieve. Every product projection $P \times Q \to Q$ is both proper and smooth. A proper map has upward-closed strict image; a smooth map has downward-closed strict image. A map that is both has image a union of connected components of the underlying comparability graph. The empty-source map satisfies both lift conditions vacuously.*

Proof. For an inclusion, an outgoing lift from p to $q \geq p$ exists exactly when $q \in P$; when it exists its lifting poset is the singleton $\{q\}$. This is precisely the cosieve condition. The dual calculation gives the sieve condition. For a product projection, the outgoing lift poset from (p, q) to q' is $\{p' \geq p\} \times \{q'\}$, with initial object (p, q'); the incoming one has terminal object (p, q'). For a general map, nonemptiness of the relevant lifting categories proves the asserted closure of its strict image. An upward- and downward-closed subset is constant along every edge of the comparability graph, hence along its connected components. If the domain is empty, there is no fixed upstairs object at which to test a lift; the conditions are vacuous and its Kan extensions have the prescribed empty-limit or empty-colimit values. Thus no unstated surjectivity on π_0 is present. □

D.8 (Proved). The precise constructible base-change statement and a geometric class where it is literal. *Under the finite-poset constructible-coefficient dictionary established in Part A, intrinsic restriction and its two adjoints satisfy D.4. This statement holds for any coefficient theory covered by that dictionary, and formally for the restriction of any derivator to finite posets. For a pointed theory the open–closed functors are those of D.1–D.2. For geometric coefficients covered by exodromy, the closed-cosieve inclusion is the literal closed inclusion $BB \hookrightarrow BX$; its right Kan extension is ordinary closed direct image. A sieve is represented geometrically by the open subset $BX \setminus BB$, with its constructible equivalence to BA, rather than by the generally non-open inclusion $BA \hookrightarrow BX$. Product projections of finite posets give a further class of literal proper, locally trivial stratified projections. For spaces, and for compactly generated or stable presentable coefficients covered by Part A, their right direct image and proper base change agree with D.4. Arbitrary monotone maps do not support an unqualified identification of intrinsic right Kan extension with ambient topological direct image.*

Proof. Transport of an adjunction through the specified coefficient equivalences identifies its left and right adjoints uniquely, including their units, counits, and Beck–Chevalley mates. D.4 therefore gives the claimed intrinsic theorem, not just an abstract equality of its two sides. For closed cosieves the realization is the closed subcomplex BB. Ambient extension by terminal objects away from this closed subspace has exactly the support characterization of i_* and preserves the given constructibility; hence it is that right adjoint. For a sieve the open complement is the one calculated in C.2. Its deformation retraction onto BA preserves the induced stratification, and the exodromy equivalence identifies its constructible coefficients with those on A. The geometric open restriction and its adjoints are consequently the specified intrinsic functors under this identification.

For product projections the natural homeomorphism $B(P \times Q) \cong BP \times BQ$ carries the minimum-vertex stratification to the product stratification. Since P is finite, the fibre BP is compact. Nonabelian proper base change applies because these polyhedra are locally compact Hausdorff (Lurie, 2009, Corollary 7.3.1.18); for the stated larger coefficient class we use Haine (2022, Corollary 3.2, Proposition 3.8, and Subexample 3.15). It identifies restriction of direct image to a target stratum with direct image of the restricted product family. Exodromy on that stratum, which is contractible, identifies the family with a constant diagram of constructible coefficients on BP; its sections are the limit over P^{op}. Thus direct image is constructible. Since ambient pullback restricts to the intrinsic restriction functor, uniqueness of its right adjoint identifies this direct image and its mate with the intrinsic ones in D.4. Base change of the projection is again the same product projection. The underlying realization square is a genuine topological pullback for finite posets: nerves preserve limits, and realization preserves them on underlying sets; a continuous bijection from the compact realized pullback to the Hausdorff pullback is a homeomorphism (Kerodon, 2026b, Theorem 3.6.2.1). This establishes a precise geometric class without calling every categorical smooth map a differential submersion.

For the final warning use $f : e \to [1]$ selecting 0. In the minimum-vertex stratification the open stratum of $B[1] = [0, 1]$ is $[0, 1)$, and its closed stratum is $\{1\}$. For a two-point space M, intrinsic f_*M is the constant diagram $M \xrightarrow{1} M$. Ambient topological direct image along $|f|$ is instead the skyscraper at 0: its stalk is M at 0 and terminal at points $0 < t < 1$. It is not locally constant on the open stratum. Therefore ambient direct image need not even preserve this constructible subcategory. Likewise, an intrinsic left adjoint $f_!$ is not, by its notation alone, a compact-support direct image. These distinctions are necessary corrections to the unrestricted six-operation reading of Lead 4. □

D.9 (Proved). The identified transcription and terminology corrections. *At I ms. 54 the target of the fibre inclusion defining smoothness must be $y_0\backslash X$, not the printed $y_0\backslash Y$. The properness criterion of I ms. 53 uses outgoing lifts, and the smoothness criterion uses incoming lifts. At XII ms. 59 the displayed closure relation is incompatible with the simultaneously specified downset topology: with downsets open, $x \leq y$ implies $y \in \overline{\{x\}}$, not $x \in \overline{\{y\}}$.*

Proof. An object of the fibre X_{y_0} maps to (x, id_{y_0}), whose second datum has target $f(x)$; it is an object of $y_0\backslash X$. Mapping it into $y_0\backslash Y$ instead loses the upstairs object and cannot have the stated comma categories of lifts. The next displayed slice-fibre criterion in the manuscript has the correct incoming variance, which confirms the correction mathematically without attributing a scanning error. The outgoing criterion is verified in D.3. For the topological assertion, the closed subsets of a poset with downsets open are upsets, so the closure of $\{x\}$ is $\{y : x \leq y\}$. Taking the two-element poset already refutes the reverse formula. The same manuscript paragraph later gives this correct upward-closure description; hence its initial specialization-order formula and its open-set convention cannot both be retained. We retain the explicit sieve convention required throughout this paper. □

E Bifibrations and the audit of VIc-C008

We use "bifibration" exclusively for an ordinary Grothendieck fibration which is also an ordinary Grothendieck opfibration. A Thomason bifibration is a different notion. All weak equivalences in this section are in W_∞. In the replacement argument, "weak fibration" means universal preservation of W_∞ under strict pullback, the condition of XI Theorem 1(a); for W_∞ this agrees with the weak-fibration tests identified in Part D.

E.1 (Proved). The universal properties required by a strict bifibration. *Let $p : X \to S$ be a Grothendieck fibration, with a cleavage and transport $\alpha^* : X_t \to X_s$ for $\alpha : s \to t$. Then p is an opfibration if and only if each α^* has a left adjoint $\alpha_! : X_s \to X_t$. In that event $B\alpha_!$ and $B\alpha^*$ are mutually inverse homotopy equivalences. Every strict bifibration is both W_∞-proper and W_∞-smooth; the converse for a specified functor is false.*

Proof. For $x \in X_s$ and $y \in X_t$, cartesian factorization identifies arrows $x \to y$ above α with $\mathrm{Hom}_{X_s}(x, \alpha^* y)$. A cocartesian arrow out of x above α represents this functor of y, hence is exactly the datum of $\alpha_! x$ and the adjunction bijection

$$\mathrm{Hom}_{X_t}(\alpha_! x, y) \cong \mathrm{Hom}_{X_s}(x, \alpha^* y).$$

The representing universal properties construct the action on arrows. They also give the full cocartesian property, not only factorization against arrows lying over α. Indeed, for $\beta : t \to u$ and $z \in X_u$, the cleavage comparison $(\beta\alpha)^* \cong \alpha^*\beta^*$ and the adjunction identify

$$\mathrm{Hom}_X^{\beta\alpha}(x, z) \cong \mathrm{Hom}_{X_s}(x, \alpha^*\beta^* z) \cong \mathrm{Hom}_{X_t}(\alpha_! x, \beta^* z) \cong \mathrm{Hom}_X^{\beta}(\alpha_! x, z).$$

Here the superscript specifies the image in S, and the composite bijection is composition with the arrow defined by the adjunction unit. This proves its universal property for every further base arrow. Uniqueness then supplies the comparison isomorphisms for composites. Conversely cocartesian factorization gives the first displayed adjunction bijections. The unit and counit induce homotopies from the identity to $B\alpha^* B\alpha_!$ and from $B\alpha_! B\alpha^*$ to the identity, respectively. Thus no additional weak-equivalence condition on transports is needed in the question once actual bifibrancy is known. Cartesian lifts give terminal objects in the incoming lifting categories; cocartesian lifts give initial objects in the outgoing lifting categories. The lifting-category tests of Part D therefore imply smoothness and properness. The finite diamond of D.6 disproves the converse. In particular the strict condition is stronger, rather than incomparable, as a property of a given functor. □

E.2 (Known). What Cisinski's factorization actually supplies. *Every functor $p : X \to S$ has a functorial factorization $X \xrightarrow{i} R_S X \xrightarrow{r} S$, with $i \in W_\infty$ and r strongly proper and smooth. If p is a locally constant Grothendieck fibration, i is a weak equivalence on every strict fibre, and r represents the same locally constant object. This statement asserts no ordinary cartesian or cocartesian universal lifting property for r.*

Proof. We opened Cisinski (2006, Theorem 5.3.14, pp. 230–231) before using it. Its cofibrantly generated model structure has the indicated fibrations, so a chosen small-object factorization is functorial. Proposition 5.3.13 gives properness and smoothness, and Corollary 6.4.8 gives local constancy. For the additional fibre assertion, the strict fibres of a Grothendieck fibration with weak-equivalence transports model the homotopy fibres of Bp by Quillen's Theorem B, with arrows reversed. The same holds for the proper and smooth map r by Cisinski (2006, Corollaries 6.4.8 and 6.4.16). The map Bi is a weak equivalence over BS. Naturality of the homotopy-fibre comparison therefore gives weak equivalences $B(X_s) \to B((R_S X)_s)$ for every

s. This includes empty fibres and components of S outside the image of p; a homotopy fibre there is empty on both sides. The specified map, not an unspecified isomorphism of total homotopy types, proves the fibrewise conclusion. □

E.3 (Proved). The exact output of the alternating replacement. *Let Φ be the comma-category Grothendieck-fibration replacement over S, and let Ψ be its opposite. The sequence of VI ms. 192–195,*

$$X = X_0 \longrightarrow X_1 = \Phi X \longrightarrow X_2 = \Psi\Phi X \longrightarrow X_3 = \Phi\Psi\Phi X \longrightarrow \cdots,$$

has a functorial colimit $R_S^{\mathrm{alt}}X$ which is proper and smooth over S. Its unit is a total weak equivalence. For locally constant weak-fibration representatives, including locally constant Grothendieck fibrations, the unit is a fibrewise weak equivalence. These assertions do not imply that the colimit is a strict bifibration.

Proof. Explicitly, $\Phi_S X$ has objects $(s, x, \alpha : s \to p(x))$ and arrows $(u, v) : (s, x, \alpha) \to (s', x', \alpha')$ with $p(v)\alpha = \alpha' u$. Projection to s is a split fibration. The map i_X sends x to $(p(x), x, \mathrm{id})$; forgetting s retracts it, and the arrows (α, id_x) give the natural transformation from the identity of $\Phi_S X$ to the resulting composite. Thus i_X is a total weak equivalence. The definition and argument for Ψ reverse all arrows.

Suppose p is a weak fibration. The point inclusion $s \to s\backslash S$ is a weak equivalence over S, since id_s is initial. Its pullback gives $X_s \to s\backslash X = (\Phi_S X)_s$ in W_∞. For $\alpha : s \to t$, precomposition gives a weak equivalence $t\backslash S \to s\backslash S$ over S, because both source and target have initial objects. Pulling it back proves that the transports $t\backslash X \to s\backslash X$ are weak equivalences. A Grothendieck fibration with these transports is again a weak fibration: arbitrary strict base change is a Grothendieck fibration with the same transports, and Quillen's Theorem B identifies all strict fibres with homotopy fibres, so pulling back a weak equivalence of bases induces a weak equivalence of total spaces. The argument for Ψ uses S/s, with terminal object id_s, and the opposite formulation of Theorem B. These are exactly the stagewise and fibrewise assertions required in the iteration. A locally constant Grothendieck fibration is an allowable initial input by the same application of Theorem B.

For completeness, the colimit step is independent of any universal lifting property. Suppose a filtered system of proper functors is given. An object, a base arrow, an arrow between two lifts, or an equation between such arrows involves finitely many data. The corresponding lifting category in the colimit is consequently the filtered colimit of the lifting categories at sufficiently late stages. The nerve preserves these filtered colimits degreewise. Filtered colimits of weakly contractible simplicial sets are weakly contractible: a map from a finite sphere, a finite homotopy between such maps, and the required finite horn fillers occur at a finite stage after Kan replacement, which commutes with filtered colimits. Thus the outgoing lifting test is retained. The incoming test is dual. The odd stages are smooth, the positive even stages are proper, and both subsequences are cofinal. Their common colimit is therefore proper and smooth.

The same finite-data argument identifies each strict fibre of the colimit with the colimit of strict fibres. Filtered colimits preserve weak equivalences of simplicial sets, so the stagewise comparisons prove the two stated claims about the unit. Naturality and two-out-of-three also show that the replacement preserves fibrewise weak equivalences on the stated subcategory. This justifies the fine-localization comparison with perfect objects, but not a comparison with strict bifibrations. The next statement exhibits the precise universal-property issue. □

E.4 (Refuted). Filtered colimits of strict bifibrations under arbitrary maps. *The claim that a filtered colimit in* $\mathbf{Cat}/S$ *of strict bifibrations is a strict bifibration is false, already for* $S = [1]$.

Proof. For $n \geq 0$, let X_n be the poset $0 < 1 < \cdots < n < t$, and let $p_n : X_n \to [1]$ send $0, \ldots, n$ to 0 and t to 1. Identities are the only endomorphisms, an arrow exists precisely for an inequality, and composition is transitivity. Its fibres are $[n]$ and e. The covariant transport $[n] \to e$ has right adjoint selecting n, so p_n is a bifibration by E.1. Include X_n in X_{n+1} by the identity on its objects. These are functors over [1]. The colimit has fibre $\mathbf{N}$ over 0 and e over 1, with one arrow $k \to t$ for every natural number k.

A cartesian lift of $0 \to 1$ with target t would require a largest natural number. Given a candidate $k \to t$, the arrow $k + 1 \to t$ cannot factor through it. Hence the colimit is not a fibration. The old cartesian arrow $n \to t$ has already lost its universal property in X_{n+1}. Nevertheless every incoming and outgoing lifting category in the colimit is contractible: the only nontrivial incoming one is $\mathbf{N}$, which has initial object 0. The colimit remains smooth and proper, exactly as E.3 requires. The code file `E_finite_checks.py` verifies the finite stages and the failure to preserve their cartesian lifts. The unbounded absence of a maximum is the proof of failure in the colimit. □

E.5 (Proved). The classifying space of the contractible-patch category. *Let* Y *be a small simplicial set and let* λ *be an infinite cardinal in the fixed universe, with* $|Y| \leq \lambda$. *Let* $K_\lambda(Y)$ *be the category of nonempty weakly contractible simplicial sets of size at most* λ *mapped to* Y, *with all strict simplicial maps over* Y. *Use fixed subset encodings so this category is small. There is a natural zigzag of weak homotopy equivalences*

$$BK_\lambda(Y) \longleftarrow B(\Delta/Y) \longrightarrow |Y|.$$

In particular the patch construction in CATHOT D.1 has the correct classifying space; Kan fibrancy is unnecessary for this assertion.

Proof. The functor $j_Y : \Delta/Y \to K_\lambda(Y)$ sends a simplex $\sigma : \Delta^n \to Y$ to that same map, and sends a simplex operator to its map of representables. Every Δ^n is nonempty and contractible and has at most countably many simplices, so it is allowed. For a patch $a : P \to Y$, the comma category $j_Y/(P, a)$ is isomorphic to Δ/P: an object is a simplex $\tau : \Delta^n \to P$, and its composite $a\tau$ specifies the required object of Δ/Y uniquely. Arrows on both sides are the same commuting triangles of simplices. Its classifying space has the weak homotopy type of P, hence is contractible. Quillen (1973, §1, Theorem A) implies $Bj_Y \in W_\infty$. The standard category-of-simplices comparison $B(\Delta/Y) \to |Y|$ is natural and a weak equivalence. On a chain of simplex maps ending in $\sigma : \Delta^n \to Y$, it sends barycentric coordinates to the corresponding affine combination of the images of the barycentres of the earlier simplices, then applies $|\sigma|$. Its weak-equivalence property follows by realizing the simplicial replacement of the diagram of standard simplices over Y: the map to $B(\Delta/Y)$ collapses its contractible values, and the augmentation to Y contracts the factorization category of each simplex using its identity factorization. This fixes the naturality needed below. For $Y = \varnothing$ both categories are empty. Cardinal coding gives at most 2^λ objects and arrows and stays inside the fixed universe. □

E.6 (Proved). What patchwise integration represents. *Let $F : S \to \mathbf{sSet}$ be a strict functor taking arrows to weak equivalences, with values bounded in size by λ. The Grothendieck construction $\int_S K_\lambda(F)$ is an opfibration and represents the covariant local system F; equivalently its fibre transport represents the corresponding contravariant local system after inversion in BS. If every transition $F(\alpha)$ is a trivial Kan fibration, this opfibration is also a fibration. For a contravariant $F : S^{\mathrm{op}} \to \mathbf{sSet}$ with trivial-Kan-fibration transitions, the fibration obtained by integrating $s \mapsto K_\lambda(F_s)^{\mathrm{op}}$ is a bifibration representing F.*

Proof. Postcomposition makes $K_\lambda(F)$ a strict functor. The fibre over s is $K_\lambda(F_s)$ and the cocartesian transport along $\alpha : s \to t$ is postcomposition with $F(\alpha)$. E.5 is a natural diagramwise comparison, so this fibre diagram is objectwise equivalent to F. Integrating the diagram gives its specified local system; the ordinary Grothendieck construction models its homotopy colimit by Thomason (1979, Theorem 1.2), with the explicit comparison checked in Cegarra (2020, §3.1, equation (9)), after reversing the variance. This is more than a comparison of two unrelated total spaces.

Suppose $f : Y \to Z$ is a trivial Kan fibration. Pullback sends a contractible patch $Q \to Z$ to $Q \times_Z Y \to Y$. The projection to Q is a trivial Kan fibration, so the new total simplicial set is nonempty and weakly contractible. The standard slice adjunction restricts to

$$f_! : K_\lambda(Y) \rightleftarrows K_\lambda(Z) : f^*.$$

Finite products and pullbacks of sets of size at most the infinite cardinal λ retain that bound. If subset codes are used, select isomorphic coded pullbacks; the adjunction is unchanged. E.1, in its covariant formulation, proves bifibrancy. For a contravariant diagram, taking opposite patch categories turns each $f_!$ into a right adjoint, which has the left adjoint required by E.1. The natural homeomorphism $B(C^{\mathrm{op}}) \cong BC$, obtained by reversing simplex coordinates, preserves the diagramwise comparison of E.5. This proves the contravariant assertion with its necessary variance. E.10 below supplies the transition hypothesis for a substantial class of bases. □

E.7 (Refuted). Weak-equivalence patch transports need not have adjoints. *Even for a weak equivalence between contractible Kan complexes, postcomposition $K_\lambda(Y) \to K_\lambda(Z)$ can have neither a left nor a right adjoint. Merely enlarging the collection of contractible patches, while retaining strict maps over the target and the same postcomposition functor, does not remedy this example.*

Proof. Let G be the groupoid on objects $0, 1$ with exactly one arrow between every ordered pair. Composition is the unique arrow with the specified endpoints. Both objects are terminal, so NG is a contractible Kan complex. Let $f : \Delta^0 \to NG$ select 0. This is a weak equivalence. Every object in the image of $K_\lambda(f)$ has constant structure map with value 0. Let $q : \Delta^0 \to NG$ select 1. For every nonempty patch P over 0, there is no simplicial map $f_!P \to q$ over NG, because such a map would equate the distinct zero-simplices 0 and 1. There is also no map $q \to f_!P$ over NG.

If $f_!$ had a right adjoint R, the identity of $R(q)$ would, by the adjunction, produce a map $f_!R(q) \to q$, contradiction. If it had a left adjoint L, the identity of $L(q)$ would produce $q \to f_!L(q)$, contradiction. Adding more nonempty contractible domains does not change either zero-simplex argument. This refutes the raw proposed construction, not the existence of a different strict-bifibration representative of its local system. Indeed this particular local system is constant and has an evident product representative. The finite groupoid and the two incompatible point patches are checked in `E_finite_checks.py`. □

E.8 (Proved). All locally constant 1-types have strict bifibration models. *For every small S, a locally constant object of* $\mathrm{HOT}(S)$ *whose values are 1-truncated spaces has a functorial strict-bifibration representative, in the sense of a functorial construction from a strict simplicial presheaf model.*

Proof. Choose a strict presheaf $F : S^{\mathrm{op}} \to \mathbf{sSet}$ with Kan values representing the object. Apply the fundamental groupoid functor objectwise. The natural maps $F_s \to N\Pi_1 F_s$ are weak equivalences because the values are 1-types. Every transition $\Pi_1 F_t \to \Pi_1 F_s$ is an equivalence of ordinary groupoids, hence has a left adjoint. The ordinary Grothendieck construction $\int_S \Pi_1 F$ is consequently a fibration and an opfibration by E.1. Its comparison with F is objectwise and natural, proving the claim in $\mathrm{HOT}(S)$. The construction of the underlying category and projection is functorial in F; choosing particular adjoints is unnecessary for the property of being a bifibration. Empty fibres cause no exception: equivalences preserve their emptiness, and the unique functor between two empty categories has itself as both adjoints. □

E.9 (Proved). All bases with 1-truncated classifying space. *If BS is a 1-type, every locally constant object of* $\mathrm{HOT}(S)$*, with arbitrary space-valued fibres, has a strict-bifibration representative. This includes groupoids, bases with an initial or terminal object, and every weakly contractible base.*

Proof. Let $G = S[\mathrm{Mor}(S)^{-1}]$ be the ordinary groupoid completion, and let $q : S \to G$ be its canonical functor. The canonical map $BS \to BG$ induces the fundamental-groupoid equivalence; because BS is a 1-type it is a weak equivalence. Restriction along Bq is therefore an equivalence of local-system infinity-categories. Represent the resulting object on G by a strict diagram $H : G^{\mathrm{op}} \to \mathbf{sSet}$ with Kan values, using the ordinary projective model for diagrams of spaces. Since every arrow of G is invertible, every structure map of this strict diagram is a simplicial isomorphism. Choose one cardinal bounding the small family of its values. Then $K_\lambda(H)$ is a diagram of categories with isomorphism transitions. Its Grothendieck construction is a bifibration and models H by E.5. Pulling that bifibration back along the specified q gives a bifibration over S, and its diagramwise class is q^*H, which is the original local system. This proof uses an equivalence induced by a specified weak-equivalence functor and a diagramwise comparison; it does not infer a fine equivalence from equality of total homotopy types.

The construction is natural after the standard choices of diagram models and replacement functors, but the arbitrary choice of an inverse to restriction is not asserted to be a literal canonical functor on the ordinary homotopy category. Over an initial or terminal base the companion's comparison with a constant diagram gives the more elementary functorial construction directly. □

E.10 (Proved). Bifibration representatives over every poset. *Let S be a small category in which every arrow is a monomorphism, or every arrow is an epimorphism. Then every locally constant object of* $\mathrm{HOT}(S)$*, with arbitrary space-valued fibres, has a small ordinary strict-bifibration representative. In particular this holds for every small poset, without a finiteness or dimension restriction. The construction is functorial on strict simplicial models after a uniform size bound for their chosen injective fibrant replacements has been fixed, and has a specified natural diagramwise weak-equivalence comparison.*

Proof. First suppose every arrow of S is a monomorphism. Represent the local system by a strict simplicial presheaf F. The injective model structure on $\mathbf{sSet}^{S^{\mathrm{op}}}$ has objectwise weak equivalences and objectwise monomorphisms as cofibrations. Its existence and functorial

factorizations follow from Lurie (2009, Proposition A.2.8.2); its compatibility with the ordinary simplicial enrichment is stated in Lurie (2009, Proposition A.3.3.2 and Remark A.3.3.4). Choose a functorial injective fibrant replacement $F \to RF$.

For $\alpha : s \to t$, let $h_s = S(-, s)$ and $h_t = S(-, t)$, regarded as discrete simplicial presheaves. Since α is a monomorphism, $h_s \to h_t$ is objectwise injective. The simplicial model axiom therefore makes

$$RF_t = \mathrm{Map}(h_t, RF) \longrightarrow \mathrm{Map}(h_s, RF) = RF_s \tag{E.10a}$$

a Kan fibration. We spell out its lifting argument. The adjoint of a horn-lifting problem for (E.10a) has left map

$$(h_t \times \Lambda^n_k) \amalg_{h_s \times \Lambda^n_k} (h_s \times \Delta^n) \longrightarrow h_t \times \Delta^n.$$

At each object this is a coproduct of identities and horn inclusions, because $h_s \to h_t$ is injective. It is consequently an objectwise trivial cofibration, against which the injectively fibrant RF has the right lifting property. The same argument with the empty presheaf in place of h_s shows that each RF_t is Kan. Naturality of $F \to RF$ and two-out-of-three show that (E.10a) is also a weak equivalence. Thus every transition of RF is a trivial Kan fibration.

Choose an infinite cardinal λ bounding all values of RF. The fixed universe is closed under this small family and under the replacement construction, so such a cardinal exists within it. E.6 now gives the bifibration

$$\int_S \big(s \longmapsto K_\lambda(RF_s)^{\mathrm{op}}\big) \longrightarrow S. \tag{E.10b}$$

The comparison $F \to RF$, the natural simplex-category zigzag of E.5, and natural reversal of nerves give a specified equivalence of its fibre diagram with F. All maps and compositions are the actual postcomposition maps of the strict diagram RF, so their coherences are retained. For any small family of input models and maps we can choose one λ bounding all replacement values; with that choice (E.10b) is functorial. We do not assert that one small cardinal bounds the values of every small presheaf simultaneously.

If every arrow of S is an epimorphism, use a strict covariant model of the corresponding local system with transports inverted in BS. The covariant representables $S(t, -) \to S(s, -)$ associated to $\alpha : s \to t$ are now objectwise injective. The same lifting proof makes all transitions of an injective fibrant replacement trivial Kan fibrations, and the covariant construction of E.6 applies. Every poset satisfies both cancellation hypotheses, proving the asserted unrestricted poset case.

In particular the six-element poset $P = \{a_0, a_1\} < \{b_0, b_1\} < \{c_0, c_1\}$ has $BP \cong S^0 * S^0 * S^0 \cong S^2$, and the path local system classified by $* \to BP$, with fibre ΩS^2, has the bifibration representative (E.10b). The program checks its six vertices, twelve edges and eight triangles. This example is affirmative, not a remaining gap. □

E.11 (Open). The remaining strict-bifibration problem. *For arbitrary small S and arbitrary locally constant $F \in \mathrm{Fun}(S^{\mathrm{op}}, \mathcal{S})$, does there exist an ordinary small strict bifibration $p : X \to S$ whose fibre diagram represents F? If so, can one choose it functorially with a specified fibrewise-equivalence comparison from a standard representative? Essential surjectivity outside E.8–E.10, and the stronger general equivalence after fibrewise localization asked in VIc-C008, remain unproved in this audit.*

An explicit remaining test has only one object. Let

$$M = \{1\} \amalg \{e_{ij} : i, j \in \{0, 1\}\}, \qquad e_{ij} e_{k\ell} = e_{i\ell}, \tag{E.11a}$$

with 1 as identity, and regard M as a one-object category. Every nonidentity element is idempotent, so none is monic or epic. The unnumbered lemma of Fiedorowicz (2002, pp. 33–34)

proves $BM \simeq S^2$ for exactly this monoid. We record its elementary verification. Its universal group is trivial since all generators are idempotent. Put $M_i = \{e_{i0}, e_{i1}\}$. The right modules $\mathbb{Z}[M_i]$ are projective, being the images of idempotent left multiplication on $\mathbb{Z}[M]$. There is an exact resolution

$$0 \longrightarrow \mathbb{Z}[M_0] \oplus \mathbb{Z}[M_1] \xrightarrow{\text{incl}} \mathbb{Z}[M] \xrightarrow{e_{00}-e_{01}} \mathbb{Z}[M_0] \xrightarrow{\epsilon} \mathbb{Z} \longrightarrow 0.$$

The middle map vanishes on all four nonidentity basis elements and sends 1 to $e_{00} - e_{01}$, proving exactness directly. Tensoring with the trivial left module gives $\mathbb{Z}^2 \xrightarrow{(1,1)} \mathbb{Z} \xrightarrow{0} \mathbb{Z}$. Thus the integral homology is that of S^2; simple connectivity, Hurewicz, and the homological Whitehead theorem give $BM \simeq S^2$. The code checks the multiplication, module maps, and these integral chain calculations.

The precise unsettled object is the path local system classified by $* \to BM$, with fibre ΩS^2. We ask for a strict bifibration over this one-object, five-arrow category representing that class and having weakly contractible total category. One object is already minimal for a nonempty base; we make no claim that five is minimal among possible numbers of arrows. All poset bases, including the six-element sphere poset, have been settled by E.10.

The attempted replacements fail at identified steps. E.2 produces homotopical lifting categories, not universal lifts. E.3 does not preserve universal arrows, as E.4 demonstrates. E.7 refutes the assertion that postcomposition on the original patch categories always has the adjoints needed to integrate in both directions; E.10 succeeds by first replacing the whole diagram under its cancellation hypotheses. For (E.11a), the representable maps used in that argument fail to be monomorphisms, so the injective lifting argument does not establish the required transition fibrations. Algebraic Kan fibrations are a possible further modification, but a left adjoint to change of base must also be proved to preserve the contractible-total-space subcategory; preservation does not follow from the existence of algebraic fillers. We do not use that unverified step.

We also searched for and opened a newer free-bifibration construction. Clarke, Scherer, and Zeilberger (2026, Theorem 1.17 and Propositions 1.23–1.24) construct a free bifibration on any ordinary functor and a fully faithful unit. Their theorem is a categorical universal property; it does not assert that the unit is a nerve weak equivalence or a fibrewise weak equivalence on locally constant input. That missing homotopical assertion is not supplied here. Lamarche (2013, §4) explicitly distinguishes his path-based weak equivalences from the standard nerve weak equivalences. We therefore do not convert the free construction into a solution by changing the localizer.

E.12 (Proved). What Chapters XI and XVI settle for this audit. *The lifting conditions of XI §1 and the factorization arguments of XVI §4 concern universal preservation of weak equivalences, perfect maps, and locally constant weak fibrations. They do not, by their stated conclusions, settle the ordinary strict-bifibration problem E.11. Two type corrections needed when reading those arguments are $f_s : X'_s \to X_s$ in XI ms. 131 and $p \in \mathcal{F}$ in the factorization $f = pi$ of XVI Lemma 4.4.1, ms. 41–42.*

Proof. XI Theorem 1, ms. 127–131, defines a Hot-fibration by stability of weak equivalences under every strict base change, or by the corresponding fibre/comma comparisons after all base changes. No clause specifies a universal cartesian arrow or universal cocartesian arrow. Its Theorem 2 compares $X' \to X$ over S, so restriction gives precisely $X'_s \to X_s$. The other typed expression has the wrong domain and codomain for the displayed map.

In XVI Lemma 4.4.1 the output class in a factorization $f = pi$ must contain p. Lemma 4.4.3(3) repeats it with this type. The subsequent discussion, ms. 42–43, first factors through a perfect map and then applies Φ to obtain a locally constant Grothendieck fibration.

This second replacement supplies only the fibration direction. The fork on VI ms. 191–192 shows that it can destroy properness, hence cannot silently be declared a bifibration. The comparison of fibration axioms in XVI Theorem 4.4.13 is therefore compatible with E.2 and E.3 and leaves the extra universal-property question distinct. □

F Stable pointed rational curves

Throughout this section the ground field is $\mathbb{C}$, the markings are individually labelled, and $n \geq 3$. We write $X_n = \overline{M}_{0,n}(\mathbb{C})$ and $U_n = M_{0,n}(\mathbb{C})$. A stratum means a stratum for the stable labelled dual tree, not a stratum in a chosen triangulation. We distinguish the exit category C_n from the entrance category $T_{0,n} = C_n^{\mathrm{op}}$. Thus arrows of C_n smooth nodes, whereas arrows of $T_{0,n}$ specialize curves. All groups below are discrete groups.

F.1 The four-pointed case

F.1 (Known). The four-pointed moduli space. *The cross ratio identifies X_4 with $\mathbb{P}^1(\mathbb{C})$ and U_4 with $\mathbb{P}^1(\mathbb{C}) \setminus \{0, 1, \infty\}$. The other three strata are points, indexed by the partitions 12|34, 13|24, and 14|23. Writing $G_4 = \langle a, b \rangle$ for the free group of rank two, we can choose positive peripheral elements*

$$m_0 = a, \qquad m_1 = b, \qquad m_\infty = (ab)^{-1}.$$

Each peripheral homomorphism $\mathbb{Z} \longrightarrow G_4$, $1 \mapsto m_i$, is injective.

Proof. An ordered triple of distinct points of $\mathbb{P}^1$ has a unique projective transformation to $0, 1, \infty$; the fourth point is the cross ratio. Stable reduction at the three excluded values gives exactly the three two-component curves with two markings on each component. This also identifies the compactification; see Keel (1992, Catalogue, Facts 1–2, pp. 551–552). Removing disjoint small discs around the three points leaves a pair of pants, which retracts to a wedge of two circles. The boundary relation, with connecting paths fixed, is $m_0 m_1 m_\infty = 1$. Each displayed word is nontrivial in a free group, which is torsion-free, proving injectivity. □

F.2 (Proved). All the arrows of C_4. *A skeleton of C_4 has objects u, p_0, p_1, p_∞, automorphism groups $\mathrm{Aut}(u) = G_4$ and $\mathrm{Aut}(p_i) = 1$, and the only noninvertible hom-sets are*

$$\mathrm{Hom}_{C_4}(p_i, u) = G_4 / \langle m_i \rangle.$$

The quotient consists of left cosets $g\langle m_i \rangle$, formed by the right action of $\langle m_i \rangle$. Postcomposition is the left G_4-action, $h \circ [g] = [hg]$. There are no arrows between distinct p_i, nor from u to a p_i. The opposite of this specified category is $T_{0,4}$.

Proof. The endomorphism space at u is the loop space of the aspherical three-punctured sphere, hence is equivalent to the discrete group G_4. The endomorphism space at a point stratum is contractible. The exit mapping space from p_i to u is the homotopy fibre over u of the punctured-disc link $S^1 \longrightarrow U_4$; radial reparametrization identifies the usual homotopy-link model with this fibre. Its components are the cosets of the image of $\pi_1(S^1)$, and each component is contractible: the long exact sequence of the homotopy fibre and the injectivity in F.1 give vanishing of all positive homotopy groups. Transporting the endpoint around $h \in G_4$ gives the displayed left action. The frontier order forbids all remaining noninvertible arrows. Alternatively the identification with the exit infinity-category follows by restricting the transporter construction in F.7. These calculations specify every hom-space and composition, so they identify the entire infinity-category, rather than merely its homotopy category. □

F.3 (Refuted). Aspherical strata alone do not imply an ordinary exit category. *The assertion that every conically stratified space with aspherical strata has an ordinary exit infinity-category is false.*

Proof. Stratify S^2 by a point p and its complement $\mathbb{R}^2$. Both strata are contractible. The exit mapping space from p to an interior point is the homotopy fibre of the link map $S^1 \longrightarrow \mathbb{R}^2$, hence is S^1, and is not discrete. An ordinary category has discrete mapping spaces. The nonzero first homology of the link is also checked from a three-edge simplicial circle in `code/F_checks.py`. Thus F.2 and F.7 require information about links and their maps, in addition to asphericity of strata. □

F.2 The five-pointed case

F.4 (Proved). The strata and groups for five markings. *There are twenty-six strata in X_5: one open stratum u, ten one-dimensional strata d_I indexed by two-element subsets $I \subset \{1,2,3,4,5\}$, and fifteen point strata $q_{I,J}$ indexed by unordered pairs of disjoint two-element subsets. Here d_I represents the split $I|I^c$, and $q_{I,J}$ has end components carrying I and J, with the remaining marking on the middle component. Each d_I is a three-punctured sphere. The groups of the corresponding exit objects are*

$$\mathrm{Aut}(u) = G_5 = \pi_1\{(x,y) \in (\mathbb{C} \setminus \{0,1\})^2 : x \neq y\}, \quad \mathrm{Aut}(d_I) = H_I \cong F_2, \quad \mathrm{Aut}(q_{I,J}) = 1.$$

An entirely group-theoretic specification of the first group is

$$G_5 = \ker(B_4 \longrightarrow \mathfrak{S}_4)/\langle(\sigma_1\sigma_2\sigma_3)^4\rangle,$$

where $B_4 = \langle \sigma_1, \sigma_2, \sigma_3 \mid \sigma_i\sigma_{i+1}\sigma_i = \sigma_{i+1}\sigma_i\sigma_{i+1},\ \sigma_1\sigma_3 = \sigma_3\sigma_1\rangle$ and the map sends σ_i to $(i, i+1)$.

Proof. A stable five-leaf tree has zero, one, or two internal edges. A single edge gives a $2|3$ partition, hence ten possibilities. A trivalent tree has two disjoint cherries, hence $\binom{5}{2}\binom{3}{2}/2 = 15$ possibilities. Gluing the normalized components identifies the open one-edge stratum with $M_{0,3} \times M_{0,4}$ and the two-edge stratum with $M^3_{0,3}$. No automorphism of a labelled stable genus-zero curve survives, since it fixes three special points on each normalized component. Normalizing the first three markings to $0, 1, \infty$ gives the displayed configuration space. Alternatively, fix the fifth marking at infinity. The configuration space of the other four points in $\mathbb{C}$, divided by the free affine action, gives U_5. The affine group is homotopy equivalent to a circle, whose loop rotates all four points once and represents the central full twist in $P_4 = \ker(B_4 \to \mathfrak{S}_4)$. Normalization of two of the four points trivializes the affine bundle, so its homotopy sequence gives exactly the quotient displayed above. The asphericity proof in F.6 identifies these fundamental groups with the full homotopy types of the strata. The finite enumeration is independently reproduced in `code/F_checks.py`. □

F.5 (Known). A complete specification of C_5. *Fix an oriented five-marked sphere S. Let $G = G_5$ act on isotopy classes of essential multicurves, by orientation-preserving homeomorphisms fixing every marking. Choose one multicurve σ_s for every stratum s of F.4: $\sigma_u = \varnothing$, $\sigma_{d_I} = \{c_I\}$, and $\sigma_q = \{c_{q,I}, c_{q,J}\}$. Put $\Delta_s = \langle t_c : c \in \sigma_s\rangle \subset G$, where t_c is a positive Dehn twist. Then every hom-set is specified by*

$$\mathrm{Hom}_{C_5}(s,t) = \{g \in G : g\sigma_s \supseteq \sigma_t\}/\Delta_s, \qquad [h] \circ [g] = [hg]. \tag{F.1}$$

In particular

$$\mathrm{Hom}(d_I, u) = G/\langle t_{c_I}\rangle, \qquad \mathrm{Hom}(q, u) = G/\Delta_q, \qquad \Delta_q \cong \mathbb{Z}^2.$$

If I is one of the two cherries of q, then

$$\mathrm{Hom}(q, d_I) \cong H_I/\langle m_{q,I}\rangle, \tag{F.2}$$

where the three $m_{q,I}$, as q ranges over the three points in $\overline{d}_I$, are the three peripheral elements of $H_I \cong \langle a_I, b_I\rangle$, namely $a_I, b_I, (a_I b_I)^{-1}$ after choices of connecting paths. These hom-sets are empty for all other pairs of distinct strata. Formula (F.1) identifies this ordinary category with the exit infinity-category of X_5.

Proof. The identification with exit paths is the case $(g, n) = (0, 5)$ of Jansen (2023, Construction 3.18 and Corollary 6.6). We verify all the displayed algebraic data. Cutting S along a multicurve gives the exact sequence

$$1 \longrightarrow \Delta_s \longrightarrow \mathrm{Stab}_G(\sigma_s) \longrightarrow \prod_v \pi_1(M_{0,\mathrm{val}(v)}) \longrightarrow 1. \tag{F.3}$$

Indeed a homeomorphism of the cut, punctured components lifts across the annuli; the ambiguity consists precisely of twists of those annuli. Distinct disjoint essential curves have independent twists, as can be checked by arcs transverse to one curve and disjoint from all the others. In genus zero with fixed leaf labels the dual tree has no automorphisms, so there is no additional quotient or permutation group. Equation (F.3) gives $\mathrm{Stab}(c_I)/\langle t_{c_I}\rangle = H_I$ and $\mathrm{Stab}(\sigma_q) = \Delta_q$.

For a fixed incident pair (q, I) choose $r_{q,I} \in G$ carrying $c_{q,I}$ to c_I. Every transporter from q to d_I is then $k r_{q,I}$ with $k \in \mathrm{Stab}(c_I)$. The conjugate $r_{q,I}\Delta_q r_{q,I}^{-1}$ is generated by t_{c_I} and the twist about the remaining curve in the four-pointed component. Modulo $\langle t_{c_I}\rangle$ the latter is its peripheral element $m_{q,I}$. This proves (F.2), including the left action of H_I. The split labels show that no other transporters exist. Finally the composition in (F.1) is well-defined: replacing g by $g\delta$, with $\delta \in \Delta_s$, changes the product on the right by δ; replacing h by $h\epsilon$, with $\epsilon \in \Delta_t$, changes hg by right multiplication by $g^{-1}\epsilon g \in \Delta_s$, because $g\sigma_s \supseteq \sigma_t$. Associativity is inherited from G. This also checks every bimodule action in the specification. □

F.6 (Proved). Composition through the one-dimensional strata, and the asphericity of every genus-zero stratum. *In the coordinates of F.5 the remaining composition is*

$$\big(G/\langle t_{c_I}\rangle\big) \times \big(H_I/\langle m_{q,I}\rangle\big) \longrightarrow G/\Delta_q, \quad ([g], [h]) \longmapsto [g\widetilde{h} r_{q,I}], \tag{F.4}$$

where $\widetilde{h} \in \mathrm{Stab}(c_I)$ lifts h. Together with multiplication and the bimodule actions, this gives all compositions in C_5. More generally the stratum of a stable labelled tree τ is exactly

$$U_\tau \cong \prod_{v \in V(\tau)} M_{0,\mathrm{val}(v)}(\mathbb{C}), \tag{F.5}$$

and each factor is aspherical.

Proof. Changing the lift $\widetilde{h}$ by a power of t_{c_I} changes its product with $r_{q,I}$ by an element of Δ_q; changing h by a power of $m_{q,I}$ does the same by the second generator of $r_{q,I}\Delta_q r_{q,I}^{-1}$. Changing g by a power of t_{c_I} is harmless for the same reason. Thus (F.4) is independent of representatives and is precisely (F.1). A nonidentity composable chain between distinct strata has length at most two, so nothing remains to specify. Different routes through d_I and d_J agree exactly

when their products determine the same coset $g\Delta_q$. We have not silently chosen simultaneous representatives c_I for which all compatible pairs are disjoint; the transporters $r_{q,I}$ retain the necessary change-of-marking information.

For (F.5), normalize the curve and label the inverse images of nodes by the half-edges of τ. This defines a map from the stratum to the displayed product. Gluing the two labelled points of each edge gives its inverse. A leaf-fixing automorphism of a stable tree fixes every vertex: each vertex is the median of three leaves in distinct incident branches. Hence the inverse constructions introduce no finite quotient. This proves the product assertion, including the absence of stack isotropy. It is also the genus-zero specialization of the gluing description in Keel (1992, Catalogue, Fact 2).

The normalization of three markings identifies $M_{0,k}$ with the ordered configuration space of $k-3$ points in $\mathbb{C}\setminus\{0,1\}$. Forgetting the last point is a locally trivial bundle with fibre a finitely punctured plane; the bundle is Fadell and Neuwirth (1962, Theorem 3). For completeness, disjoint small discs about a configuration and isotopies supported in these discs give local trivializations by moving the punctures. The punctured plane retracts to a graph, and its universal cover is a tree. Induction on k, starting from $M_{0,3} = *$, and the homotopy exact sequence of this bundle show that $\pi_i(M_{0,k}) = 0$ for $i \geq 2$. Products preserve this property. Thus the asphericity assertion includes $k = 3$ and every stable tree, without invoking compactified moduli asphericity. □

F.3 The category for arbitrary numbers of markings

F.7 (Known). The ordinary exit category and the Charney–Lee model. *Let $S_{0,n}$ be an oriented n-punctured sphere and $\Gamma_n = \pi_0 \operatorname{Homeo}^+(S^2; 1, \dots, n)$, with markings fixed pointwise. Let $\mathcal{C}_n$ be the set of isotopy classes of finite essential multicurves, including the empty one. Define a small ordinary category C_n by*

$$\operatorname{Ob}(C_n) = \mathcal{C}_n, \qquad \operatorname{Hom}_{C_n}(\sigma, \tau) = \{g \in \Gamma_n : g\sigma \supseteq \tau\}/\Delta_\sigma, \qquad [h] \circ [g] = [hg]. \tag{F.6}$$

The quotient is by the right action of Δ_σ, so its elements are left cosets $g\Delta_\sigma$. Then $NC_n \simeq \operatorname{Exit}(X_n)$ as infinity-categories over the poset of dual trees. The category is EI and its automorphism group at σ is $\prod_v \Gamma_{\mathrm{val}(v)}$. Isomorphism classes of objects are stable labelled dual trees. The category $T_{0,n} = C_n^{\mathrm{op}}$ is equivalent to the original Charney–Lee category of stable labelled topological curves and isotopy classes of degenerations.

Proof. The exit equivalence is Jansen (2023, Corollary 6.6); the identification with the original Charney–Lee category is that paper's Proposition 3.23. These results establish an equivalence of infinity-categories, not merely a weak equivalence of classifying spaces. Their hypotheses apply because X_n is the analytification of the stable-curve moduli stack, and in labelled genus zero every stabilizer of an algebraic curve is trivial. The boundary is a simple normal-crossings divisor by Keel (1992, Catalogue, Fact 1). The definition and the typing of composition have been verified in F.5; that proof uses no restriction to five markings. If $g\sigma \supseteq \sigma$, equality follows from finiteness and equal cardinality; consequently every endomorphism is invertible. Equation (F.3) proves the asserted automorphism group in every valence. Smoothing annuli identifies the orbits of multicurves with labelled stable trees, and the converse follows by gluing homeomorphisms of the normalized components across annuli. This proves the remaining algebraic assertions. A skeleton is finite on objects but generally infinite on arrows. We retain the nonskeletal model (F.6) when making claims about strict pullbacks of functors. □

F.8 (Proved). Coefficients and the necessary opposite. *For a compactly generated presentable infinity-category $\mathcal{C}$ there is an equivalence*

$$\mathrm{Sh}_{\mathrm{cbl}}(X_n;\mathcal{C}) \simeq \mathrm{Fun}(C_n,\mathcal{C}) = \mathrm{Fun}(T_{0,n}^{\mathrm{op}},\mathcal{C}). \tag{F.7}$$

In particular $\mathrm{HOT}(T_{0,n}) \simeq \mathrm{Ho}\,\mathrm{Sh}_{\mathrm{cbl}}(X_n;\mathcal{S})$. *If $\mathbb{D} = \mathbb{D}_{\mathcal{C}}$ is the presheaf-variance derivator of such a $\mathcal{C}$, the same assertion holds at every diagram shape. For a bare derivator, $\mathbb{D}(T_{0,n})$ and its shifted values define a theory of categorical constructible coefficients; no equivalence with a separately defined category of topological $\mathbb{D}$-valued sheaves follows from Der 1–4 alone.*

Proof. Equation (F.7) is the coefficient conclusion of Jansen (2023, Theorem 6.7). Apply it also to the diagram coefficient category $\mathrm{Fun}(I^{\mathrm{op}},\mathcal{C})$, or use the exponential law, to obtain the assertion at shape I. Taking homotopy categories gives the displayed derivator statements. The last sentence separates a definition from a theorem: an abstract derivator supplies the categories $\mathbb{D}(A)$ and homotopy Kan extensions, but does not supply sheaves on the topological site of X_n. Nor does the cited coefficient theorem assert its conclusion for every presentable $\mathcal{C}$ without an additional coefficient hypothesis. Finally, the prompt's literal combination "$T_{0,n} = \mathrm{Exit}(X_n)$" and "$\mathrm{HOT}(T_{0,n}) =$ constructible sheaves", with the fixed presheaf variance, requires an opposite. Our notation makes that correction explicit rather than changing the direction of exits. □

F.4 Clutching, forgetting markings, and base change

F.9 (Proved). The actual functor that forgets a marking. *Filling the puncture $n+1$ induces a homomorphism $\phi : \Gamma_{n+1} \to \Gamma_n$ and a function $q : \mathcal{C}_{n+1} \to \mathcal{C}_n$ which discards newly peripheral curves and replaces newly isotopic curves by a single isotopy class. The assignments*

$$Q : C_{n+1} \longrightarrow C_n, \quad \sigma \longmapsto q\sigma, \quad [g] \longmapsto [\phi(g)] \tag{F.8}$$

define a strict functor. Its opposite $F : T_{0,n+1} \to T_{0,n}$ represents the geometric morphism $f : X_{n+1} \to X_n$ which forgets the last marking and stabilizes. Repeated forgetting and relabelling are compatible with composition; on chosen skeletons the compatibility is by specified natural isomorphisms arising from the chosen markings.

Proof. The operation on curves is independent of isotopy, preserves inclusion, and is equivariant for ϕ. A twist becomes the twist about the image essential curve or the identity if the image is peripheral. Thus $\phi(\Delta_\sigma) \subseteq \Delta_{q\sigma}$ and the coset map is well-defined. If $g\sigma \supseteq \tau$, then $\phi(g)q\sigma \supseteq q\tau$, which checks the source and target of every arrow. The formula respects multiplication, identities, and composition. Deleting two punctures successively and reducing the curve systems gives the same set of surviving essential isotopy classes as deleting both at once, so the functors on the fixed marked-surface models compose strictly.

For a permutation of the labels choose an orientation-preserving homeomorphism r inducing it. Relabelling acts on C_n by $\sigma \mapsto r\sigma$ and $[g] \mapsto [rgr^{-1}]$. If another lift is $r' = ar$ with $a \in \Gamma_n$, the arrows $[a] : r\sigma \to r'\sigma$ are a natural isomorphism: both composites in its naturality square are represented by $argr^{-1}$. For two permutations compare their product lift with the chosen lift of the product by this same formula. Associativity of multiplication gives the coherence identity. Thus no splitting of the full mapping class group onto the permutation group is assumed. Filling a relabelled puncture commutes with these assignments because filling is equivariant on curve classes and homeomorphisms.

A pinching map of marked surfaces commutes with filling a puncture followed by contraction of its unstable component: on the complement of the pinched annuli this is the same continuous

map; on an unstable sphere it is the canonical collapse. Twisting a pinched annulus is precisely the equivalence already divided out in (F.6). Therefore the induced functor on degeneration categories is (F.8) with its direction reversed. Equivalently, the maps on strata and on the normal annuli give (F.8) under exit paths. This supplies the comparison with the geometric map, not just an abstract functor with the correct object map. □

F.10 (Proved). Properness of the specified forgetful functor. *For the full multicurve models (F.6), the entrance functor $F : T_{0,n+1} \to T_{0,n}$ is W_∞-proper in Grothendieck's sense. Thus its right-Kan base-change transformation is invertible for every strict base-change square and every derivator in the conventions of this paper. The assertion also holds for repeated forgetting of markings. It is an assertion about these specified functors, not about arbitrary replacements by equivalent skeletal categories.*

Proof. We use the outgoing-arrow lifting criterion of Part D, or its incoming-arrow dual on exit categories. Fix $\sigma \in \mathcal{C}_{n+1}$, put $\tau = q\sigma$, and fix an arrow $\beta : \eta \to \tau$ of C_n. The relevant category has objects arrows $a : \rho \to \sigma$ in C_{n+1} with $q\rho = \eta$ and $Q(a) = \beta$; its morphisms are triangles commuting over η. If $\beta = [b]$, choose $\gamma \in \Gamma_{n+1}$ with $\phi(\gamma) = b$. The assignments $(\rho, [g]) \mapsto (\rho, [\gamma^{-1}g])$ and the identity on triangle morphisms give an isomorphism to the lifting category with target $\gamma^{-1}\sigma$ and base arrow $[1] : \eta \to b^{-1}\tau$; the inverse multiplies by γ. Thus, after this specified comparison, we can write the test as $\beta = [1] : \eta \to \tau$ with $\eta \supseteq \tau$. Surjectivity is the Birman exact sequence

$$1 \longrightarrow \pi_1(S_{0,n}) \longrightarrow \Gamma_{n+1} \xrightarrow{\phi} \Gamma_n \longrightarrow 1; \tag{F.9}$$

for $n = 3$ it is also immediate from F.1. For $n \geq 4$ we use Kent IV, Leininger, and Schleimer (2009, §2.1).

An object is now $(\rho, [g])$ with $g\rho \supseteq \sigma$ and $\phi(g) \in \Delta_\eta$. The arrow $[g] : \rho \to g\rho$ is an isomorphism over η and identifies this object with $(g\rho, [1])$. The full subcategory of objects $(\rho, [1])$ is exactly the poset

$$P(\eta; \sigma) = \{\rho \in \mathcal{C}_{n+1} : \rho \supseteq \sigma, q\rho = \eta\}, \quad \rho \leq \rho' \iff \rho \supseteq \rho'. \tag{F.10}$$

Indeed a triangle between two such objects is represented by h with $[h] = [1] \in \mathrm{Hom}(\rho, \sigma)$, hence $h \in \Delta_\rho$. It therefore exists precisely when $\rho \supseteq \rho'$ and is then unique. This proves an equivalence of categories with (F.10), including its arrows, rather than only a bijection on connected components.

We prove that (F.10) has contractible nerve. First suppose $\sigma = \varnothing$. Realize η by disjoint geodesics in a complete hyperbolic metric on the n-punctured sphere. In its universal disc their preimages are disjoint proper lines. The graph dual to these lines is a tree: it is connected, and every line separates the disc, so the corresponding edge separates the graph. The vertices are components of the complement, and the edges are the lifted lines. The deck group $K = \pi_1(S_{0,n})$ acts with vertex stabilizers $\pi_1(V)$ for the components V of $S_{0,n} \setminus \eta$ and edge stabilizers the cyclic subgroups generated by the curves of η.

Point-pushing identifies lifts of η having the same number of curves with the vertices of this tree. In detail, moving the forgotten puncture into a component V gives a lift; two moves give the same isotopy class exactly when their difference is a loop in V, so these lifts are the cosets $K/\pi_1(V)$. A lift with one extra curve whose two images are isotopic places the forgotten puncture between two parallel curves. Such lifts are the cosets of the corresponding edge subgroup and are incident to the two lifts obtained by deleting one of the parallel curves. This is precisely the barycentric incidence description of the dual tree. This description is also the content of Kent IV et al. (2009, Theorem 7.1); their Lemma 2.1 checks that at most one duplication occurs.

There is one additional possibility because (F.10) allows curves that become peripheral: a disc contains the forgotten puncture and exactly one retained puncture. Such a lift adds one peripheral leaf to the regular lift in the adjacent component. Its stabilizer under point-pushing is the cyclic group of that puncture. It has a unique proper face still mapping to η, obtained by deleting this extra curve. There cannot be two such peripheral curves, nor a peripheral curve and a duplicated pair, because only one puncture is filled. Thus the nerve is a subdivided tree with leaves attached. It is contractible. When $\eta = \varnothing$ there is a central vertex $\varnothing$ and only these leaves, so the same conclusion holds without using a nonempty curve complex.

For general σ, cut the pointed sphere along σ before filling the puncture. Every component except the one containing the forgotten puncture has a unique lift of each prescribed further refinement in $\eta \setminus \tau$: isotopy classes there are unchanged by filling a puncture on another component. If the distinguished component remains stable after filling, the remaining choices are exactly the preceding tree-and-leaves calculation on that component, whose other special points are treated as punctures. If it becomes unstable, it had exactly three special points before filling. They are either the forgotten marking, a retained marking, and a node, or the forgotten marking and two nodes. In the first case it is a fixed peripheral leaf; in the second it is a fixed inserted edge. Preserving σ fixes this placement, and the additional curves of η lift uniquely in the other components. In either case (F.10) is a singleton. These cases exhaust the possibilities. We have proved contractibility of every required lifting category, and hence properness. Proper functors are closed under composition, which proves the final assertion. □

F.11 (Refuted). Forgetting a marking need not be smooth. *The proper entrance functor $T_{0,5} \to T_{0,4}$ is not W_∞-smooth. The first stable forgetful map $T_{0,4} \to T_{0,3} = e$ is both proper and smooth, so five markings are the smallest instance of this failure in the genus-zero forgetful sequence.*

Proof. Take the upstairs point stratum $x = q_{12,34}$, with marking 5 on the middle three-pointed component. Its two curves c_1, c_2 become the same essential curve c after forgetting 5. Write $\Delta_x = \langle t_1, t_2 \rangle \cong \mathbb{Z}^2$ and $\phi(t_1) = \phi(t_2) = t_c$. In exit variance consider the arrow $[1] : qx = \{c\} \to \varnothing$. Its outgoing lifting category is the test for smoothness in entrance variance.

The target of any such lift has empty image under q. It is therefore either the empty curve system or a single curve enclosing 5 and one retained marking. The latter cannot be a subset of any translate of σ_x: its split label is different from both 12|345 and 34|125, and pure mapping classes preserve split labels. Hence the only possible target is the open object. Put $K = \ker(\phi)$. Objects of the lifting category are $\{g\Delta_x : \phi(g) \in \langle t_c \rangle\}$, and arrows between them are postcomposition by elements of K. Because $\phi(\Delta_x) = \langle t_c \rangle$, this is the transitive action groupoid of K on $K/(K \cap \Delta_x)$. Its stabilizer is

$$K \cap \Delta_x = \langle t_1 t_2^{-1} \rangle \cong \mathbb{Z}.$$

The lifting category therefore has classifying space $B\mathbb{Z}$, not a contractible space. This verifies every hypothesis and the failure of the smoothness criterion. Independence of t_1, t_2 follows from their disjoint essential supports as in F.5; in particular the stabilizer is not killed by the quotient defining the arrows. The check of the exponent map $(r, s) \mapsto r + s$ is included in the code, although the counterexample category itself is infinite. A functor to the terminal category passes both lifting tests because its tests are ordinary slices with an initial or terminal object. This settles the minimality assertion within the sequence. □

F.12 (Proved). Clutching as a closed categorical embedding. *Let $L = A \amalg B$ be a partition with $|A|, |B| \geq 2$. Clutching the extra marked points gives an isomorphism*

$$\overline{M}_{0,A\cup\{*\}} \times \overline{M}_{0,B\cup\{*\}} \xrightarrow{\sim} D_{A|B} \subset X_{|L|} \tag{F.11}$$

onto the closed boundary divisor. In entrance variance the induced functor can be represented by the fully faithful cosieve on the objects whose dual trees contain the split $A|B$. It is therefore W_∞-proper. On exit categories the corresponding inclusion is a fully faithful sieve. The product description and this inclusion are compatible with clutching more than one node. The inclusion of the fixed-curve product model into the full multicurve model is not itself W_∞-proper; the proper representative is its replete enlargement just specified.

Proof. Normalization at the distinguished node inverts (F.11); fixed labels distinguish the two branches, including when their cardinalities are equal. This is Keel (1992, Catalogue, Fact 2). Choose a fixed curve c separating A from B, and first use only multicurves containing c. An arrow between two such objects is represented by a mapping class preserving c: in the image of the source multicurve there is exactly one curve with the labelled split $A|B$, so it must be the fixed target curve. The stabilizer of c modulo its twist is the product of the mapping class groups of the two punctured components. This proves the product identification of categories and their full faithfulness.

For the actual cosieve take the full replete subcategory of all multicurves containing some curve with split $A|B$. It is equivalent to the fixed-c product just described, and it is a sieve in exit variance since refinement retains the split. Its opposite is the required literal cosieve. This distinction matters: the inclusion of the fixed-c subcategory alone is not replete in the nonskeletal model. In fact choose a pure mapping class taking c to a distinct curve c'; such a class exists by twisting about an essential curve intersecting c, since both sides of c contain at least two markings. The entrance isomorphism $\{c\} \to \{c'\}$ has no lift under the fixed-c inclusion: its target object $\{c'\}$ is not in that inclusion's literal image. Its outgoing-arrow lifting category is empty, proving the claimed failure of strict properness for that particular representative. Repeating the normalization argument at several disjoint nodes gives the stated compatibility. The properness of the replete full cosieve follows from the lifting criterion of Part D. □

F.13 (Proved). Strict categorical fibres and geometric fibres of forgetting. *For the full model (F.8), the strict fibre $F^{-1}(\tau)$ is equivalent to the entrance category of the stable n-pointed curve represented by τ, stratified by its smooth component complements, its marked points, and its nodes. This is the topological fibre of $X_{n+1} \to X_n$. In particular the fibre of $X_5 \to X_4$ is S^2 over U_4 and $S^2 \vee S^2$ over any point stratum; the map is not a topological fibre bundle and is not a submersion of the underlying smooth manifolds.*

Proof. An object upstairs that stabilizes to a fixed nodal curve places the extra point either on the smooth locus of a component, at a retained marking by adding a three-pointed bubble, or at a node by inserting a three-pointed bridge. These are exactly the strata of that curve. In the strict categorical fibre, two placements of the same type are isomorphic by point-pushing. At a smooth-component placement the automorphism group is the kernel of forgetting in its component mapping class group, namely the fundamental group of that component with its special points removed, by (F.9). At a bubble or bridge the automorphism group is trivial. The incidence bimodule at each adjacent marked point or node is the set of left cosets of the corresponding peripheral cyclic subgroup in that component group, by the same annular calculation as F.2. There are no chains through three different strata. This identifies every

hom-set, action, and composition with the entrance category of the curve, and proves the claimed categorical equivalence.

The geometric assertion about the fibre is the universal-curve interpretation of forgetting and stabilizing; it is also described in Keel (1992, Introduction, pp. 546–548). A smooth four-pointed curve is one projective line. A boundary four-pointed curve is two projective lines meeting in one point. Their second integral homology has rank one and two respectively. Fibres of a topological fibre bundle over a connected base are homeomorphic and have equal homology. A proper smooth submersion would be a fibre bundle, and in this instance the singular fibre also directly excludes a submersion. This geometric argument is independent of, and weaker than, the categorical obstruction in F.11. □

F.14 (Proved). The coefficient functors and their base-change properties. *For a compactly generated presentable coefficient infinity-category, pullback along clutching or forgetting corresponds under (F.7) to restriction along the entrance functors above. Their ordinary topological direct images preserve constructibility and correspond to right Kan extension. For every derivator, the specified proper categorical models satisfy right-Kan base change for all strict category pullbacks. Geometric proper base change holds for the analytic maps and their pullbacks among locally compact Hausdorff spaces. The two assertions agree whenever the strict pullback category is identified with the entrance category of the geometric pullback, in particular for the point fibres in F.13 and for the closed-boundary squares in F.12. There is no smooth-base-change assertion for forgetting: F.11 refutes that hypothesis already at five markings.*

Proof. The geometric maps preserve strata and send exit paths to exit paths, so the pullback assertion follows from exodromy and F.9–F.12. For a closed inclusion, direct image is extension from the closed subset, and its values and gluing maps are the corresponding right Kan extension; the local open–closed calculation is the one in Part D. Forgetting is a proper algebraic morphism. Its restriction over a fixed target stratum is locally topologically trivial as a family of stratified nodal curves: choose disjoint coordinate discs around the labelled special points on each normalized component, move the points by isotopies in these discs, and glue the fixed node pairs. Shrink to a contractible coordinate ball B in the target stratum, and write its stratified trivialization as $B \times C$, with C the stratified nodal fibre. By the definition of exit simplices, the exit category of this product is $\Pi_\infty(B) \times \mathrm{Exit}(C) \simeq \mathrm{Exit}(C)$. Applying coefficient exodromy to the product shows that the constructible sheaf on it is pulled back from C, up to an equivalence compatible with restriction to each fibre. This proves the required locally constant transport; local triviality of the underlying spaces alone would not supply that conclusion for arbitrary sheaves. Compactness of the fibre and proper base change identify the stalk of the direct image with sections on this fibre. They therefore make the direct image locally constant on the target stratum. This proves preservation of constructibility.

Since both inverse image and direct image preserve the respective constructible subcategories, their adjunction restricts to those subcategories. Under (F.7), inverse image is restriction. Uniqueness of its right adjoint therefore identifies the restricted direct image with right Kan extension, including its unit and counit. In addition, F.10 computes right Kan extension at τ as the homotopy limit over the strict fibre, and F.13 identifies that limit with sections of the restriction to the geometric fibre. These are the specified fibrewise base-change comparisons. For spaces, geometric proper base change is Lurie (2009, Corollary 7.3.1.18); testing with compact generators extends it to the indicated coefficient categories, because mapping out of a compact generator commutes with the filtered stalk colimit and with the limits in the direct image. The categorical assertion uses only the properness theorem of Part D and hence applies also to bare derivators. For a square obtained by pulling back a replete boundary inclusion, the strict pullback is the full subcategory on objects whose image lies in that boundary. Its

geometric counterpart is the closed union of strata with exactly this condition. The exit category of a closed union of strata is the full subcategory on those strata: along an exit simplex all strata precede its last stratum, and closedness retains all these predecessors. This proves the asserted comparison for boundary squares. Nothing in this argument identifies an arbitrary strict category pullback with a geometric pullback. That comparison is extra data, and its verified instances were specified in the statement. Likewise the left adjoint of constructible pullback is a left Kan extension; it should not be confused with the topological exceptional direct image, whose adjunction is with exceptional inverse image. □

F.15 (Proved). The explicit map from five to four markings. *Forget marking* 5*. The open object maps to the open object. A divisor object* d_I *maps to the open object when* $5 \in I$ *and to the boundary partition* $I|\{1,2,3,4\} \setminus I$ *otherwise. Every* $q_{I,J}$ *maps to the partition defined by its surviving essential split; if* 5 *is the middle marking then both splits survive as the same partition. On all hom-sets the map is the coset map* $[g] \mapsto [\phi(g)]$ *of F.9. In particular a twist about a curve enclosing* 5 *and one retained mark maps to* 1*, and all other single-curve twists map to the twist about their image split.*

Proof. Deleting 5 makes the side $\{5, i\}$ unstable and removes its edge. Every other divisor split remains stable. For a two-cherry tree, deleting an endpoint marking contracts its unstable endpoint component and leaves the other split; deleting the middle marking contracts that component and identifies its two incident edges. These are all the cases. Annular twists give the arrow formulas, which are independent of chosen representatives by F.9. Over each boundary partition there are two divisor strata and five point strata, and over the open stratum there are the open stratum and four divisor strata. The program enumerates these fibres and all incidence relations, and checks that the object map is monotone. The twenty-six-object description, the explicitly defined group G_5, the coset bimodules (F.1)–(F.2), and the multiplication (F.4) thus give a complete specification of $T_{0,5} \to T_{0,4}$ after opposites and choices of skeleton markings. □

F.5 What the Teichmüller tower comparison says

F.16 (Known). The relevant definitions in Grothendieck's texts. *In* La Longue Marche*, §19, the notation is* $A_g = \mathrm{Aut}(X_g)$ *for the topological homeomorphism group of a compact orientable genus-g surface,* $A_{g,\nu} = \{u : u(S_{g,\nu}) = S_{g,\nu}\}$*,* $A^!_{g,\nu} = \{u : u|_{S_{g,\nu}} = \mathrm{id}\}$*, and*

$$\Gamma_{g,\nu} = \pi_0 A_{g,\nu}, \qquad \Gamma^!_{g,\nu} = \pi_0 A^!_{g,\nu}. \tag{F.12}$$

Our Γ_n *is the orientation-preserving subgroup of* $\Gamma^!_{0,n}$*, not the full group allowing permutations or orientation reversal. The modular topos* $M_{g,\nu}$ *of §26 is the modular topos of smooth curves of type* (g, ν)*; its complex topological fundamental group is the corresponding oriented Teichmüller group. It is not the constructible-sheaf category of the compactification. The operation called "bouchage de trous" in §27 induces the forgetful operation of F.9 on the open stratum.*

Proof. The definitions (F.12) are the literal equations (3)–(11) in the attached §19 transcription, manuscript p. 102, scans 204–205. The superscript ! imposes fixed labels; orientation preservation is an additional restriction, not part of the symbol A_g in those equations. This resolves the potential mismatch between analytic moduli with labelled markings and the whole homeomorphism group. The supplied P034 extract starts with the continuation of §26; for its preceding definition we opened the Grothendieck Circle transcription, Part Ib, §26, equations (19)–(25), printed pp. 5–6. There $M_{g,\nu,\mathbb{C}}$ is identified through its Teichmüller

universal cover, and equation (24) is the universal-curve fundamental group extension. The topological restriction of that extension is (F.9) with the same label and orientation restrictions. At §27, manuscript pp. 172–173, scans 340–342, filling the punctures in $I \setminus I'$ is explicitly the operation considered. On the surface fundamental group it kills the normal closure of those peripheral loops, by van Kampen. On homeomorphism components it is the homomorphism ϕ of F.9. This proves the asserted compatibility while making no assertion about the arithmetic extensions discussed on the surrounding pages (Grothendieck, 1981, §§19,26–27). □

F.17 (Proved). The precise comparison with the Lego groupoid. *The full subgroupoid of $T_{0,n}$ on the open stratum is equivalent to $B\Gamma_n$. More generally its maximal subgroupoid is*

$$\coprod_{\tau} B\left(\prod_{v} \Gamma_{\mathrm{val}(v)}\right). \tag{F.13}$$

The usual topological Teichmüller groupoid of U_n is equivalent to the first of these groupoids. The equivariant path groupoid formed from the marking complex of the Lego–Teichmüller game and its mapping class group action presents this groupoid after imposing the same label and orientation conventions and capping marked boundary circles to punctures. The nonequivariant path groupoid of that complex is contractible. Neither construction identifies the whole degeneration category $T_{0,n}$ with a groupoid.

Proof. The first two assertions are (F.3) and (F.6): an isomorphism cannot change the number or labelled configuration of nodes. For the opposite category we identify the opposite of each automorphism group with that group by inversion. Since U_n is connected and aspherical, its fundamental groupoid is equivalent to its one-object groupoid $B\Gamma_n$, with the equivalence depending on a basepoint and connecting paths. Grothendieck's *Esquisse*, §2, explicitly treats the system of open moduli spaces linked by filling and gluing, and advocates several basepoints for its fundamental groupoids; see the attached edition, pp. 9–11 (Grothendieck, 1984).

For a genus-zero extended surface Σ with $n \geq 3$ marked boundary circles, let $\mathcal{L}(\Sigma)$ be the two-complex of markings and elementary moves of Bakalov and Kirillov Jr. (2000, Main Theorem 4.9). That theorem states that it is connected and simply connected. Form the groupoid with objects its markings and morphisms $M \to M'$ the pairs $(g, [p])$ consisting of a mapping class g and a path $p : gM \to M'$ modulo homotopy relative to endpoints. For $q : hM' \to M''$, composition is $(h, [q]) \circ (g, [p]) = (hg, [q * (hp)])$, where hp is the translate of p and the concatenation first traverses hp and then q. Simple connectedness makes the path class unique whenever its endpoints are fixed. Consequently the functor sending $(g, [p])$ to g is an equivalence to the mapping class group's one-object groupoid. Without the group action there is exactly one path class between any pair of markings, so the path groupoid is equivalent to the terminal category. This distinction is consistent with Bakalov and Kirillov Jr. (2000, Remark 4.3): an elementary marking move is not itself a homeomorphism of surfaces. Passing from marked boundary circles to labelled punctures kills the boundary twists: extension over the punctured discs gives a surjection, and the kernel consists exactly of annular twists, by the same cutting argument as (F.3). Restricting to fixed labels gives $B\Gamma_n$. This proves the asserted relation to the Lego presentation.

For $n \geq 4$, $T_{0,n}$ has noninvertible degeneration arrows, for example $u \to p_0$ in $T_{0,4}$. Such an arrow cannot become invertible under an equivalence of categories. Thus the whole category is not a groupoid. Its infinity-groupoid completion has the homotopy type of X_n, by F.7 and the groupoid-completion property of exit paths. Already $X_4 \simeq S^2$ differs from $U_4 \simeq BF_2$. Ordinary groupoid completion would lose even the second homotopy group of S^2. Hence neither completion should be silently substituted for the open Teichmüller groupoid. The input for the separate profinite package is the specified category (F.6), its opposite convention,

the functors F.9 and F.12, and the properness and nonsmoothness in F.10–F.14. No profinite completion or arithmetic action is asserted here. □

G The limits of the dictionary

G.1 (Proved). The two fine variances have a coefficient counterpart. *The enhanced contravariant and covariant diagram theories over S are $\mathrm{Fun}(S^{\mathrm{op}}, \mathcal{S})$ and $\mathrm{Fun}(S, \mathcal{S})$; opposition of the base exchanges them. Their locally constant parts both identify with $\mathrm{Fun}(BS, \mathcal{S})$. There is no inference from this last identification to an equivalence of the two fine theories over the unchanged base, or to Verdier duality. For an EI base their abstract stratified counterparts are the entrance and exit variances of B.1–B.2; for a non-EI base B.7 obstructs the usual conservative stratified-space interpretation.*

Proof. Precomposing the first assignment with $S \mapsto S^{\mathrm{op}}$ gives the second assignment literally. Inverting all arrows in either base gives the same space, with inverse paths accounting for the variance switch. Before that localization the two assignments remember directed arrows. For the finite fork $S = (a \leftarrow b \rightarrow c)$, S has an initial object and no terminal object, whereas S^{op} has a terminal object and no initial object. Both are already idempotent complete. B.6 therefore proves that their presheaf infinity-categories are not equivalent; the same fork is checked in the program for G.2. In a stable coefficient category, a Verdier-duality construction would additionally require a dualizing object and a suitable dualizable or constructible finiteness range. Neither an operation on the indexing category nor the axioms Der 1–4 provide those data. Thus the two fine localizations discussed in VI have a precise diagrammatic counterpart, but their interchange does not by itself supply a six-operation duality theorem. □

G.2 (Proved). Total asphericity is an exactness condition on indexing. *Let A be a nonempty connected small category. The following three conditions are equivalent: A is totally W_∞-aspherical in XII ms. 88–89; every component of the category of elements of every covariant $H : A \to \mathbf{Set}$ is weakly contractible; and for every strict $K : A \to \mathbf{sSet}$ the canonical augmentation*

$$\operatorname*{hocolim}_A K \longrightarrow \operatorname*{colim}_A K \tag{G.2}$$

is a weak equivalence. This is a coefficient-indexing exactness property, not a property determined by BA alone and not a condition on the local conical charts of a stratification.

Proof. We verify the supplied TOTASPH B.11 reduction. The manuscript definition says that the category of elements of $u_!1_A$ is weakly contractible for every functor $u : A \to B$, where this Kan extension is the ordinary presheaf Kan extension. Its value at b is $\pi_0(b\backslash A)$. The functor

$$A \longrightarrow B/(u_!1_A), \qquad a \longmapsto (u(a), [\mathrm{id}_{u(a)}])$$

is ordinary final: its comma category over (b, c) is exactly the nonempty connected component c of $b\backslash A$. Conversely an ordinary final u has $u_!1_A = 1_B$. Totality is therefore equivalent to every ordinary final image of A being weakly contractible.

Assume totality. In particular BA is contractible. Given a nonempty connected component E of $\mathrm{El}_A H$, replace H by the subfunctor consisting of that component. Adjoin a new source t with arrows $t \to a$ indexed by $H(a)$; composition is the action of H. The inclusion of A is ordinary final, since its only new comma category is E. Its target is therefore weakly contractible. The one-object case of C.3 identifies its classifying space with the homotopy pushout of $BA \leftarrow BE \rightarrow *$. The reduced homology sequence shows that E is integrally acyclic.

This acyclicity property is inherited by each such E. Indeed for $L : E \to \mathbf{Set}$ the functor $a \mapsto \coprod_{h \in H(a)} L(a,h)$ has category of elements isomorphic to $\mathrm{El}_E L$, by the map $((a,h),l) \mapsto (a,(h,l))$. Fix a point of connected BE and any cyclic subgroup C of its fundamental group. The coset action of $\Pi_1(BE)$ defines a set-valued locally constant functor on E whose element category is the connected cover with fundamental group C. This can be seen directly by lifting strings of arrows: a loop closes precisely when its class lies in C. The inherited acyclicity makes its first homology $C_{\mathrm{ab}} = C$ vanish. Every cyclic subgroup is therefore trivial. Thus BE is simply connected and acyclic, hence weakly contractible by the first-nonzero-group form of Hurewicz. This proves the second condition, including categories of elements with any number of components. Conversely if the second condition holds, the commas of an ordinary final $u : A \to B$ are connected categories of elements of $a \mapsto B(b, u(a))$. They are contractible. Quillen's Theorem A gives $BA \simeq BB$, and the constant singleton diagram makes BA contractible. This proves totality.

For a simplicial diagram K, its simplicial replacement has bidegrees

$$\coprod_{a_0 \to \cdots \to a_p} K(a_0)_q.$$

At fixed q this is $N\,\mathrm{El}_A(K_q)$. Under the second condition its component map to the discrete set $\mathrm{colim}_A K_q$ is a weak equivalence. The bisimplicial diagonal lemma gives exactly (G.2). Conversely apply (G.2) to a discrete set-valued diagram. Its source is the nerve of the element category and its target its component set, so every component must be contractible. These are the actual augmentations, not abstract equivalences between their endpoints.

Finally let A be the fork $a \leftarrow b \to c$, with its three identities and two indicated arrows. It has initial object b, so BA is contractible. Set $H(b) = \{0,1\}$ and $H(a) = H(c) = \{*\}$, with both action maps constant. Its element category has the four objects b_0, b_1, a, c and four nonidentity arrows $b_i \to a, b_i \to c$, with no other nonidentity compositions. Its nerve is the four-edge circle, so the second condition fails. In contrast the point category satisfies it. The program `G_checks.py` checks both finite categories and their integral cycle ranks. Filtered categories do satisfy the criterion, because finite diagrams in an element component admit the requisite common targets and equalizations; we make no converse assumption. In particular the exactness condition in (G.2) is always stated when an ordinary colimit is used in place of a homotopy colimit. □

G.3 (Proved). Truncation localizers do have a sheaf meaning. *For $n \geq 1$, the localizer $W_n = \{u : \tau_{\leq n} Bu \text{ is an equivalence}\}$ is exactly the class detected by equivalence on locally constant sheaves of $(n-1)$-types on the CW realizations. This does not make W_n equivalences stable under arbitrary ordinary homotopy pullbacks. For every finite $n \geq 0$, the map $* \to K(\mathbb{Z}, n+1)$ lies in W_n, whereas its pullback along itself has projection $K(\mathbb{Z}, n) \to *$, which is not in W_n.*

Proof. A nerve realization is locally contractible, so its locally constant coefficients are functors on its fundamental ∞-groupoid. B.11 gives the stated test with $r = n-1$. For the counterexample, $K(\mathbb{Z}, n+1)$ is n-connected, whereas the homotopy pullback of its two basepoints is its loop space. The loop space has nonzero π_n if $n > 0$, and is the nontrivial discrete group $\mathbb{Z}$ if $n = 0$. Thus the specified comparison fails at exactly the claimed truncation level.

There is also a finite categorical $n = 0$ counterexample. Let B be the one-object group C_2, and let E be the groupoid on objects $0, 1$ with one arrow $i \to j$ for every ordered pair. Define $p : E \to B$ by sending that arrow to $j - i \pmod 2$. Addition verifies functoriality. Every base arrow has a unique lift with a fixed source or fixed target, so p is a strict bifibration. The source and target of p are connected, so $p \in W_0$. Its strict pullback along the point of B is the

discrete two-object category: only arrows with $j - i = 0$ survive. Its projection to the point is not in W_0. The program checks all compositions, both unique-lift properties, and the pullback components. Thus even a strict bifibration that models ordinary homotopy pullbacks need not preserve a larger localizer. □

G.4 (Known). The intrinsic W_n fibration criterion. *Let $p : T \to B$ be a functor, $n \geq 0$, and define a weak W_n-fibration to mean that strict pullback along p preserves every W_n-map of categories over B. This is equivalent to requiring, for every $X \to B$, that the specified map be an equivalence of n-types:*

$$\tau_{\leq n} B(T \times_B X) \longrightarrow \tau_{\leq n} BT \underset{\tau_{\leq n} BB}{\times} \tau_{\leq n} BX. \tag{G.4}$$

For $n = 0$, writing B_j for a connected component and $T_j = T \times_B B_j$, this is equivalent to the two elementary conditions that $\pi_0(T_b) \to \pi_0(T_j)$ is bijective for each $b \in B_j$, and that each arrow $b \to b'$ has a lift inside each component of T_j.

Proof. The properness hypothesis has a direct source. In Cisinski (2006, §9.2.1, pp. 364–365), W_n is the fundamental localizer generated by trivializing the single category $\Delta/\partial\Delta^{n+2}$, and it is explicitly proved proper by Corollary 4.3.28. Corollary 9.2.15 there identifies its equivalences with the maps inducing a bijection on π_0 and isomorphisms on π_i for $1 \leq i \leq n$, exactly our truncation convention. Thus its localized homotopy theory is that of n-types. The associated Thomason model is right proper by Theorem 5.2.15, and its derived pullback corresponds to the right side of (G.4), a limit of n-types. Proposition 6.2.3 of Cisinski (2006), whose proper-localizer hypothesis is essential, identifies weak fibrations precisely with functors whose every strict pullback square is a derived pullback in that theory. This proves the criterion, including the canonical comparison map. Equivalently, factor p over B into a W_n-map followed by a Thomason fibration. The weak-fibration comparison theorem makes this replacement a W_n-map on every strict pullback; the model pullback then computes (G.4). Conversely (G.4) sends a W_n-map over B to an equivalence on its right side, hence on its left side, which is precisely the defining preservation property. These are the checks required in the supplied POLES C.9 argument.

For clarity the $n = 0$ description can be proved independently. Pull back the component equivalence $e_b \to B_j$ to get the first condition. Pull back the two endpoint inclusions into the arrow category $[1] \to B$; each fibre must induce the same component bijection in the resulting two-layer category. Components can meet its two layers only through an arrow over the specified base arrow, giving the second condition. Conversely fix a connected component of X mapping into B_j. The first condition labels every fibre component by a unique $c \in \pi_0(T_j)$; the second condition connects equal labels across every edge of a zigzag in X. Distinct labels cannot connect, since their images in T_j would then connect. This gives, naturally in X,

$$\pi_0(T \times_B X) = \pi_0(T) \times_{\pi_0(B)} \pi_0(X),$$

including empty components. It proves the assertion at level zero without importing an ordinary-space homotopy-pullback claim. For $n > 0$ the right side of (G.4) is a pullback *of truncated spaces*; it cannot be replaced by truncation of an arbitrary ordinary pullback. □

G.5 (Proved). The exact universal base-change obstruction for other localizers. *For a fundamental localizer in the relative convention of 0.1, stability of its induced equivalences of spaces under every homotopy pullback holds exactly for $W = W_\infty$ and for the class of all functors. Equivalently, every W_∞-perfect categorical functor preserves W under strict pullback only in these two cases. This is the precise obstruction; it does not deny the coefficient meanings in B.11 or G.3–G.4.*

Proof. Every fundamental localizer contains W_∞, by the minimality theorem (Cisinski, 2006, Theorem 6.1.18). Model a space by its simplex category, naturally and by a weak equivalence as in E.5. Model any cospan of spaces by a strict cospan of simplicial sets, using a path object to strictify its specified homotopy, and then by their simplex categories. Factor one leg through the strongly proper and smooth replacement of E.2. Such a replacement is a weak fibration for W_∞: its strict fibres compare with commas by properness, its comma transitions are weak equivalences by smoothness and local constancy, and Quillen's Theorem B, after every base change, identifies its strict pullbacks with homotopy pullbacks. This is also the application of Cisinski's Proposition 6.2.3 in the supplied POLES E.4 proof. Consequently categorical preservation along these perfect maps is exactly preservation under all homotopy pullbacks of spaces.

Suppose this preservation holds and there is $f \in W \setminus W_\infty$. A homotopy fibre F of Bf is not weakly contractible. Its map to the point lies in W by pullback stability. If F is empty, this already gives $\varnothing \to * \in W$. Otherwise every choice of basepoint $* \to F$ is in W, by two-out-of-three. Pulling it back along itself gives $\Omega F \to * \in W$, and the argument can be iterated. If F is disconnected, it already has two points in different components. If it is connected, some homotopy group must be nonzero; otherwise it would be weakly contractible. Iterating loops produces a W-aspherical space Z with at least two components. The two basepoint maps from distinct components belong to W, but their homotopy pullback is empty. Again $\varnothing \to * \in W$. Pulling this map back to any space Y gives $\varnothing \to Y \in W$. Two-out-of-three then puts every map $X \to Y$ in W. Thus W is total. If no such f existed, minimality would give $W = W_\infty$. Conversely weak homotopy equivalences, and plainly all maps, have the stated stability.

This proof verifies POLES E.5 under the relative localizer axioms and uses no accessibility assumption. It concerns equivalences of ordinary spaces and the specified perfect replacement. The properness hypothesis on the localizer in G.4 instead computes pullbacks in its localized homotopy theory; these are different assertions, as G.3 demonstrates. □

G.6 (Proved). Accessible coefficient localizations need not be geometric sheaf localizations. *Let $L : \mathcal{S} \to \mathcal{L} \subset \mathcal{S}$ be an accessible reflective localization whose inverted maps W are preserved by product with every space. Then the constant-coefficient class of the derivator of $\mathcal{L}$ is exactly W. Accessible fundamental localizers admit this description through the simplex-category correspondence. It does not follow that L preserves finite limits or that $\mathcal{L}$ is the sheaf ∞-topos of a stratified space.*

Proof. For $B \in \mathcal{L}$ and an arbitrary space A, the exponential $\mathrm{Map}(A, B)$ is local: testing $f \in W$ against it is the same as testing $f \times A$ against B. Thus mapping spaces between local objects are local. The constant-coefficient comparison over $f : X \to Y$ is

$$\pi_0 \mathrm{Map}(Y, \mathrm{Map}(A, B)) \longrightarrow \pi_0 \mathrm{Map}(X, \mathrm{Map}(A, B)), \qquad A, B \in \mathcal{L}.$$

It is bijective whenever $f \in W$. Conversely take $A = *$, which is local, and every local B. These tests say that $\mathrm{Hom}_{\mathrm{Ho}\,\mathcal{L}}(LY, B) \to \mathrm{Hom}_{\mathrm{Ho}\,\mathcal{L}}(LX, B)$ is bijective for every B. Yoneda makes Lf an equivalence, so $f \in W$. The accessible localized category is presentable and complete, giving its derivator.

The correspondence for accessible fundamental localizers is Cisinski (2006, Theorems 4.2.15 and 1.4.3), using the test category Δ. Product stability in the categorical presentation follows from relative Theorem A: the slices of $u \times C$ over c are $u \times (C/c)$, and projection identifies them with u up to specified weak equivalences since C/c has a terminal object. This checks the product hypothesis required above and the realization argument of GALOIS E.7. No left exactness was used. If a nontrivial such reflector preserved finite limits, its equivalences would

be stable under all homotopy pullbacks, contrary to G.5 unless its class were W_∞ or total. In particular finite truncation has a useful coefficient interpretation without supplying that geometric localization property. □

G.7 (Proved). Geometric structures not supplied by the categorical data. *Neither the layered category nor the bare derivator axioms specify a conically smooth atlas, a differentiable normal bundle, a chosen tubular neighbourhood, a manifold dimension, or an orientation and dualizing object. The categorical proper and smooth conditions also do not impose a proper-support operation or a stratified submersion structure. These additional notions therefore have no automatic counterpart in the portions of the manuscript audited here.*

Proof. A one-stratum contractible manifold of any positive dimension has exit ∞-groupoid equivalent to a point and the same locally constant coefficient categories. Thus even its dimension is not determined by that categorical datum. Smooth atlases and tubular neighbourhoods include specified coordinate changes, derivatives and embeddings, none of which is supplied by a functor to a poset or an indexing derivator. An orientation or dualizing object is extra coefficient data, as the need for a specified duality in G.1 already shows. D.5–D.8 give completely specified finite counterexamples separating categorical properness and smoothness from topological properness, submersions and ambient direct image. Conically smooth stratifications do yield useful additional structure in stratified topology, but its presence must be stated and verified on the geometric side; it is not reconstructed from the names *propre* and *lisse*. □

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